

1. (c) : We have, $R = \{(f, g) : f(0) = g(1) \text{ and } f(1) = g(0)\}$

For reflexive: $(f, f) \in R$

$\Rightarrow f(0) = f(1)$ must hold $\forall f$

But $f(0) \neq f(1)$

Therefore, R is not reflexive.

For symmetric: If $(f, g) \in R \Rightarrow (g, f) \in R$

Now, $g(0) = f(1)$ and $g(1) = f(0)$, which is true $\forall f$

$\therefore R$ is symmetric

For transitive: If $(f, g) \in R$ and $(g, h) \in R$

$\Rightarrow (f, h) \in R$

Now $(f, g) \in R \Rightarrow f(0) = g(1)$ and $f(1) = g(0)$

$(g, h) \in R \Rightarrow g(0) = h(1)$ and $g(1) = h(0)$

For $(f, h) \in R$, we need $f(0) = h(1)$ and $f(1) = h(0)$

But, $f(0) = g(1) = h(0) \neq h(1)$ and $f(1) = g(0) = h(1) \neq h(0)$

Hence, R is not transitive.

2. (a) : $A = \{-3, -2, -1, 0, 1, 2, 3\}$

xRy if and only if $-2y \leq x^2 \leq 4 - 2y$

$y = -3 \quad 6 \leq x^2 \leq 10 \Rightarrow x \in \{-3, 3\}$

$y = -2 \quad 4 \leq x^2 \leq 8 \Rightarrow x \in \{-2, 2\}$

$y = -1 \quad 2 \leq x^2 \leq 6 \Rightarrow x \in \{-2, 2\}$

$y = 0 \quad 0 \leq x^2 \leq 4 \Rightarrow x \in \{-2, -1, 0, 1, 2\}$

$y = 1 \quad -2 \leq x^2 \leq 2 \Rightarrow x \in \{-1, 0, 1\}$

$y = 2 \quad -4 \leq x^2 \leq 0 \Rightarrow x \in \{0\}$

$y = 3 \quad -6 \leq x^2 \leq -2 \Rightarrow$ Value of x does not exist

$R = \{(-3, -3), (3, -3), (-2, -2), (2, -2), (-2, -1), (2, -1), (-2, 0), (-1, 0), (0, 0), (1, 0), (2, 0), (-1, 1), (0, 1), (1, 1), (0, 2)\}$

$\therefore l = 15$

To make it reflexive we will add $(-1, -1), (2, 2), (3, 3)$ in R

$\therefore l + m = 15 + 3 = 18$

3. (b) : We have, $A = \{-2, -1, 0, 1, 2, 3\}$ and

$R = \{(-2, 1), (-1, 1), (0, 1), (1, 1), (2, 2), (3, 3)\}$

Now, number of elements in R i.e., $l = 6$

For R to be reflexive,

$R = \{(-2, -2), (-1, -1), (0, 0), (-2, 1), (-1, 1), (0, 1), (1, 1), (2, 2), (3, 3)\}$

So, we need to add three elements to make it reflexive.

$\therefore m = 3$

For R to be symmetric,

$R = \{(-2, 1), (1, -2), (-1, 1), (1, -1), (0, 1), (1, 0), (1, 1), (2, 2), (3, 3)\}$

So, we need to add three elements to make it symmetric.

$\therefore n = 3$

So, $l + m + n = 6 + 3 + 3 = 12$

4. (a) : We have, $A = \{-3, -2, -1, 0, 1, 2, 3\}$, R is defined on A as xRy such that $2x - y \in \{0, 1\}$.

i.e., $2x - y = 0$ or $2x - y = 1$

$\therefore R = \{(0, 0), (-1, -2), (1, 2), (0, -1), (2, 3), (1, 1), (-1, -3)\}$ i.e., $l = 7$

For R to be reflexive, i.e., we need 5 more elements $\{(2, 2), (-1, -1), (3, 3), (-3, -3), (-2, -2)\}$ so $m = 5$ and for R to be symmetric, we need 5 more elements $\{(-2, -1), (2, 1), (-1, 0), (3, 2), (-3, -1)\}$, so $n = 5$.

$\therefore l + m + n = 7 + 5 + 5 = 17$.

5. (a) : Partitions of set $\{1, 2, 3\}$ is $\{\{1\}\}, \{2\}, \{3\}, \{\{1, 2\}, \{3\}\}, \{\{1, 3\}, \{2\}\}, \{\{2, 3\}, \{1\}\}, \{\{1, 2, 3\}\}$

$\therefore$ Number of non-empty equivalence relations on the set $\{1, 2, 3\} = 5$

6. (a) : Let $A = \{1, 2, 3, 4\}$

$R = \{(1, 2), (2, 3), (3, 3)\}$ [Given]

For an equivalence relation ' R ' should be reflexive, symmetric and transitive.

$\therefore$ Minimum elements that should be added are

$(2, 1) (3, 2) (1, 1), (2, 2), (4, 4) (1, 3) (3, 1)$

$\therefore$ Minimum number of elements required = 7.

7. (d) : $R = \{(x, y) : x, y \in \mathbb{Z} \text{ and } x + y \text{ is even}\}$

For reflexive : $x + x = 2x$ is even

For symmetric : $x + y = y + x$ is even

For transitive : $x + y$ is even and $y + z$ is even then $z + x$ is also even.

So, the relation is an equivalence.

8. (a) : $\sec^2 x - \tan^2 x = 1, \forall x \in \left[0, \frac{\pi}{2}\right)$

$\therefore R$ is reflexive

Consider, $\sec^2 x - \tan^2 y = 1$

$\Rightarrow 1 + \tan^2 x - (\sec^2 y - 1) = 1$

$\Rightarrow 1 + \tan^2 x - \sec^2 y + 1 = 1$

$\Rightarrow \sec^2 y - \tan^2 x = 1$

$\therefore R$ is symmetric.

Now, if $\sec^2 x - \tan^2 y = 1$ and $\sec^2 y - \tan^2 z = 1$

Adding both equations,

$\sec^2 x - \tan^2 y + \sec^2 y - \tan^2 z = 2$

$\Rightarrow \sec^2 x - \tan^2 z = 1$ [$\because \sec^2 y - \tan^2 y = 1$]

$\therefore R$ is transitive

Thus, R is an equivalence relation.

9. (a) : $R_1 = \{(x, y) : 2x - 3y = 2\}$

$R_1 = \{(4, 2), (7, 4), (10, 6), (13, 8), (16, 10), (19, 12)\}$

So, 6 elements are needed to make R_1 symmetric

$\Rightarrow M = 6$

$R_2 = \{(x, y) : -5x + 4y = 0\}$

$R_2 = \{(4, 5), (8, 10), (12, 15), (16, 20)\}$

So, 4 elements are needed to make R_2 symmetric

$\Rightarrow N = 4$

$\therefore M + N = 6 + 4 = 10$

10. (c) : Given, $A = \{1, 2, 3, 4, 5\}$

$R = \{(x, y) : 4x \leq 5y, x, y \in A\}$

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5), (4, 4), (4, 5), (5, 4), (5, 5)\}$

$\therefore n(R) = 16 = m$

Elements needed to make R symmetric = $\{(2, 1), (3, 1), (4, 1), (5, 1), (3, 2), (4, 2), (5, 2), (4, 3), (5, 3)\}$ i.e., 9 elements

$\therefore n = 9$

So, $m + n = 16 + 9 = 25$

11. (b) : We have, $(a, b) R (c, d)$

$\Rightarrow 3ad - 7bc$ is an even integer.

For reflexive : $(a, b) R (a, b) \Rightarrow 3ab - 7ab = -4ab$, which is an even integer.

For symmetric, $(a, b) R (c, d) \Rightarrow 3ad - 7bc$ is an even integer.

$\Rightarrow 3ad - 7bc + 4bc + 4ad$ is also an even integer ($\because$ even + even = even number)

$\Rightarrow 7ad - 3bc$ is an even integer

$\Rightarrow 3cb - 7da$ is also an even integer

$\Rightarrow (c, d) R (a, b)$

For transitive, $(a, b) R (c, d)$ and $(c, d) R (e, f) \Rightarrow 3ad - 7bc$ and $3cf - 7de$ is an even integer.

For $a = 2, b = 5, c = 6, d = 8, e = 9, f = 1$

$3af - 7bc = 3 \times 2 \times 1 - 7 \times 5 \times 9 = 6 - 315 = -309$, which is not an even integer.

$\therefore$ Given relation is not transitive.

12. (c) : Consider, $(x, x) \in R_1$

Then, $x^2 + x^2 \neq 1$ for any $x \in \mathbb{R}$

$\therefore R_1$ is not reflexive

Hence, R_1 is not an equivalence relation.

For R_2

(i) Let $(x, y) R (x, y) \Rightarrow x + y = y + x$, which is true for all $x, y \in \mathbb{N}$.

$\therefore R_2$ is reflexive.

(ii) Let $(x, y) R (z, p) \Rightarrow x + p = y + z$

$\Rightarrow p + x = z + y$

$\Rightarrow z + y = p + x$

$\Rightarrow (z, p) R (x, y)$

$\therefore R_2$ is symmetric.

(iii) Let $(a, b) R (c, d)$

$\Rightarrow a + d = b + c$... (i)

$(c, d) R (e, f) \Rightarrow c + f = d + e$... (ii)

From (i) and (ii), we have $a + d + c + f = b + c + d + e$

$a + f = b + e \Rightarrow (a, b) R (e, f)$

$\therefore R_2$ is transitive.

So, R_2 is an equivalence relation.

13. (d) : Given, $A \in M \Rightarrow A \cap A = \phi$,
if $A = \phi$
So, R is not reflexive.

If $(A, B) \in R \Rightarrow A \cap B \neq \phi$

$\Rightarrow B \cap A \neq \phi \Rightarrow (B, A) \in R$

So, R is symmetric.

If (A, B) and $(B, C) \in R \Rightarrow A \cap B \neq \phi$ and $B \cap C \neq \phi$

$\Rightarrow A \cap C$ is not necessarily non-empty set.

Hence, $A \cap C$ not necessarily belongs to R .

So, R is not a transitive relation.

$\therefore$ Given relation is symmetric and reflexive only.

14. (d) : Reflexive : For $(a, b) R (a, b)$

$\Rightarrow ab - ab = 0$ is divisible by 5.

So, $(a, b) R (a, b) \forall a, b \in \mathbb{Z}$

$\therefore R$ is reflexive.

Symmetric : For $(a, b) R (c, d)$, if $ad - bc$ is divisible by 5.

Then, $bc - ad$ is also divisible by 5.

So, $(c, d) R (a, b) \forall a, b, c, d \in \mathbb{Z}$

$\therefore R$ is symmetric.

Transitive : For $(a, b) R (c, d) \Rightarrow ad - bc$ is divisible by 5

and $(c, d) R (e, f) \Rightarrow cf - de$ is divisible by 5

Let $ad - bc = 5k_1$ and $cf - de = 5k_2$, where k_1 and k_2 are integers.

$\therefore adf - bcf = 5k_1f \dots(i)$

and $cfb - deb = 5k_2b \dots(ii)$

Solving (i) and (ii), we get

$adf - bcf + cfb - deb = 5k_1f + 5k_2b$

$\Rightarrow adf - deb = 5(k_1f + k_2b)$

$\Rightarrow d(af - be) = 5(k_1f + k_2b)$

$af - be$ is not divisible by 5 for $\forall a, b, c, d, e, f \in \mathbb{Z}$

So, R is not transitive.

$\therefore R$ is Reflexive and symmetric but not transitive.

15. (c) : To make R an equivalence relation, R should be reflexive, symmetric and transitive.

$\therefore R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (1, 3), (3, 1), (3, 2), (2, 3)\}$

16. (d) : Given, $A = \{1, 2, 3, 4, 5, 6, 7\}$

and $R = \{(x, y) \in A \times A : x + y = 7\}$

For Reflexive : Let $y = x$

So, $x + x = 7 \Rightarrow x = \frac{7}{2}$, which is not possible.

So, given relation is not reflexive.

For symmetric : $xRy \Rightarrow x + y = 7$

$\Rightarrow y + x = 7 \therefore xRy \Rightarrow yRx$

Hence, given relation is symmetric.

For transitive : Let xRy and $yRz \Rightarrow x + y = 7$ and $y + z = 7$

But it does not implies that, $x + z = 7$

So, R is not transitive.

17. (c) : $R = \{(a, b) : 3a - 3b + \sqrt{7} \text{ is an irrational number}\}$

Reflexive : For (a, a) ,

we have $3a - 3a + \sqrt{7} = \sqrt{7}$, which is an irrational number.

$\therefore R$ is reflexive

Symmetric : Let $(a, b) \in R$ i.e., $3a - 3b + \sqrt{7}$ is an irrational number

Now, we need to check $(b, a) \in R$ or not.

Let $3a = \sqrt{7}$ and $3b = 8$ then $3a - 3b + \sqrt{7}$

$= \sqrt{7} - 8 + \sqrt{7} = 2\sqrt{7} - 8$, which is an

irrational number but $3b - 3a + \sqrt{7}$

$= 8 - \sqrt{7} + \sqrt{7} = 8$, which is not an irrational

$\therefore$ For $(a, b) \in R, (b, a) \notin R$

$\therefore R$ is not symmetric.

Transitive : Let (a, b) and $(b, c) \in R$

Let $3a = 8, 3b = 2\sqrt{7}, 3c = \sqrt{7}$ then,

$3a - 3b + \sqrt{7} = 8 - 2\sqrt{7} + \sqrt{7} = 8 - \sqrt{7}$,

which is an irrational also,

$3b - 3c + \sqrt{7} = 2\sqrt{7} - \sqrt{7} + \sqrt{7} = 2\sqrt{7}$,

which is an irrational but

$3a - 3c + \sqrt{7} = 8 - \sqrt{7} + \sqrt{7} = 8$,

which is not an irrational $\Rightarrow (a, c) \notin R$

$\therefore R$ is not transitive.

18. (d)

19. (d) : Reflexive : $\gcd(a, a) = a \neq 1$ for $a \neq 1$.

Symmetric : Let $a = 2, b = 1$

$\gcd(a, b) = 1; b \neq 2a$

Thus, $(2, 1) \in R$

$\gcd(1, 2) = 1$. But $2 = 2 \times 1$

Therefore $(1, 2) \notin R$.

Hence, R is not symmetric. Transitive :

$(2, 3) \in R, (3, 8) \in R$. But $(2, 8) \notin R$

Hence, R is not transitive

R is neither symmetric nor transitive

20. (c) : If ' R ' be a relation defined on $\mathbb{N}$ as $a R b$ is $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$.

(i) $a R a \Rightarrow 5a$ is a multiple of 5

R is a reflexive relation.

(ii) $a R b, 2a + 3b = 5\alpha$ (let)

Now, $b R a$

$\Rightarrow 2b + 3a = 2b + \left(\frac{5\alpha - 3b}{2}\right) \cdot 3$

$= \frac{15}{2}\alpha - \frac{5}{2}b = \frac{5}{2}(3\alpha - b)$

$= \frac{5}{2}(2a + 2b - 2\alpha) = 5(a + b - \alpha)$

So, R is symmetric relation.

(iii) $a R b \Rightarrow 2a + 3b = 5\alpha$;

$b R c \Rightarrow 2b + 3c = 5\beta$

Now, $2a + 5b + 3c = 5(\alpha + \beta)$

$\Rightarrow 2a + 5b + 3c = 5(\alpha + \beta)$

or $2a + 3c = 5(\alpha + \beta - b) \Rightarrow a R c$

So, R is a transitive relation.

Hence, relation ' R ' is an equivalence relation.

21. (b) : Given relation; $R = \{(a, b), (b, c)\}$

and set = $\{a, b, c\}$

For symmetric, Since $(a, b), (b, c) \in R$

So, $(b, a), (c, b)$ be in R . For transitive, Since,

$(a, b), (b, c) \in R$

So, (a, c) should be in R . then (c, a) should also be in R since (a, c) lies in R .

Since, $(a, b), (b, a) \in R$. So, $(a, a) \in R$ should also lie in R .

Since, $(c, b), (b, c) \in R$. So, (c, c) should also lie in R .

Since, $(b, c), (c, b) \in R$. So, (b, b) should also lie in R .

$\therefore$ Elements to be added are :

$(b, a), (c, b), (a, c), (c, a), (a, a), (c, c), (b, b)$.

Total no. of elements to be added = 7.

22. (b) : We have, $(a, b) R (c, d)$

$\Leftrightarrow ad(b - c) = bc(a - d)$

$\Rightarrow \frac{b - c}{bc} = \frac{a - d}{ad}$ and

$\frac{1}{c} - \frac{1}{b} = \frac{1}{d} - \frac{1}{a} \Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{d} - \frac{1}{c}$

For reflexive : $(a, b) R (a, b) \Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{b} - \frac{1}{a}$ which is false.

Hence, it is not reflexive.

For symmetric, $(a, b) R (c, d)$

$\Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{d} - \frac{1}{c}$

$\Rightarrow \frac{1}{c} - \frac{1}{d} = \frac{1}{b} - \frac{1}{a} \Rightarrow (c, d) R (a, b)$

Hence it is symmetric.

For transitive :

$(a, b) R (c, d) \Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{d} - \frac{1}{c}$

and $(c, d) R (e, f) \Rightarrow \frac{1}{c} - \frac{1}{d} = \frac{1}{f} - \frac{1}{e}$

$\Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{e} - \frac{1}{f} \neq (a, b) R (e, f)$

Hence, it is not transitive.

23. (d) : For relation $T = a^2 - b^2 \in I$

Then, (b, a) on relation $R \Rightarrow b^2 - a^2 \in I$

$\therefore T$ is symmetric,

$S = \{(a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0\}$

$2 + \frac{a}{b} > 0 \Rightarrow \frac{a}{b} > -2 \Rightarrow \frac{b}{a} > \frac{-1}{2}$

If $(b, a) \in S$, then $2 + \frac{b}{a}$ not necessarily positive

$\therefore S$ is not symmetric.

24. (d) : $R_1 = \{(a, b) : ab \geq 0\}$

Case I : $(a, a) \in R_1$ because $a^2 \geq 0$

So, reflexive.

Case II : If $(a, b) \in R_1 \Rightarrow ab \geq 0$

So, $(b, a) \in R_1$ because $ba \geq 0$

So, R_1 is symmetric.

Case III : If $(a, b) \in R_1$ and $(b, c) \in R_1$

then $ab \geq 0, bc \geq 0$

but $ac \not\geq 0$ [Let as consider an example

$a = 2, b = 0, c = -2$ $a \cdot b \geq 0, b \cdot c \geq 0$

but $a \cdot c = -4 \not\geq 0$

$\therefore (a, c) \notin R$

So, R_1 is not transitive. R_1 is not equivalence.

In $R_2 = \{(a, b) : b \leq a\}$

$\Rightarrow (5, 4) \in R_2$ but $(4, 5) \notin R_2$

So, R_2 is not symmetric.

Hence, R_2 cannot be equivalence.

25. (d)

26. (b) : We have $R_1 = \{(a, b) \in N \times N :$

$|a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$

In R_1 : Since $|4 - 13| = 9 \leq 13$

$\therefore (4, 13) \in R_1$ and $(13, 20) \in R_1$

but $(4, 20) \notin R_1$

$\therefore R_1$ is not transitive.

Therefore R_1 is not equivalence.

In R_2 : $(26, 6) \in R_2$ and $(6, 39) \in R_2$

but $(26, 39) \notin R_2$ ($\angle |26 - 39| = 13$)

$\therefore R_2$ is not transitive.

Therefore R_2 is not equivalence.

27. (d) : $A = A_1 \cup A_2 \dots \cup A_k$

$A_i \cap A_j = \emptyset, i \neq j$

$R = \{(x, y) : y \in A_i \text{ iff } x \in A_i, 1 \leq i \leq k\}$

(i) Reflexive : $(x, x) \in R$; as $x \in A_i$ iff $x \in A_i$

$\therefore R$ is reflexive.

(ii) Symmetric : Let $(x, y) \in R$

$\Rightarrow x, y \in A_i$ i.e., $y, x \in A_i$

$\Rightarrow (y, x) \in R$

$\therefore R$ is symmetric.

(iii) Transitive : Let $(x, y) \in R$ and $(y, z) \in R$

$\Rightarrow x \in A_i$ iff $y \in A_i$ and $y \in A_i$ iff $z \in A_i$

$\Rightarrow x \in A_i$ iff $z \in A_i \Rightarrow (x, z) \in R$

$\Rightarrow R$ is transitive.

Hence, R is an equivalence relation.

28. (c) : $0 < |x - y| \leq 1$ is symmetric as

$|x - y| = |y - x|$ but not transitive.

For example : $(1, 2), (2, 1) \in R$ but

$(1, 1) \notin R$.

29. (a) : Reflexive : Let $(x, x) \in R$

$\Rightarrow x^3 - 3x^3 - x^3 + 3x^3 = 0$

$\therefore R$ is reflexive.

Symmetric : Let $(x, y) \in R$

$\Rightarrow x^3 - 3x^2y - xy^2 + 3y^3 = 0$

$\nRightarrow 3x^3 - yx^2 - 3y^2x + y^3 = 0$

So, R is not symmetric.

Similarly, R is not transitive.

30. (c) : $(a, b) \triangleleft (c, d) \Leftrightarrow ad = bc$

$\Rightarrow \frac{a}{c} = \frac{b}{d} \Rightarrow \frac{4}{c} = \frac{3}{d} \Rightarrow \frac{c}{d} = \frac{4}{3}$

$(c, d) \in (2, 3, 4, \dots, 30)$

$\therefore (c, d) = (4, 3), (8, 6), (12, 9), (16, 12),$
 $(20, 15), (24, 18), (28, 21)$

$\therefore 7$ ordered pair exist.

31. (b) : Equivalence class of $(1, -1)$ is a

circle with centre $(0, 0)$ and radius $\sqrt{2}$. i.e.,

$x^2 + y^2 = 2$

$\therefore$ Required set is $S = \{(x, y) \mid x^2 + y^2 = 2\}$

32. (a) : For R_1 : Let $a = 2 + \sqrt{2}$,

$b = 2 - \sqrt{2}$, $c = 2^{5/4}$

$\therefore aR_1b \Rightarrow a^2 + b^2$

$= (2 + \sqrt{2})^2 + (2 - \sqrt{2})^2 = 12 \in Q$

$bR_1c \Rightarrow b^2 + c^2$

$= (2 + \sqrt{2})^2 + (2 - \sqrt{2})^2 = 12 \in Q$

$aR_1c \Rightarrow a^2 + c^2 = (2 + \sqrt{2})^2 + (2^{5/4})^2$

$= 6 + 8\sqrt{2} \notin Q$

$\therefore R_1$ is not transitive

For R_2 : Let $a = 2^{3/2}$, $b = 3^{3/4}$, $c = 1$

Then, $aR_2b \Rightarrow a^2 + b^2 = (2^{3/2})^2 + (3^{3/4})^2$

$= 8 + 3\sqrt{3} \notin Q$

$bR_2c \Rightarrow b^2 + c^2 = (3^{3/4})^2 + (1)^2$

$= 3\sqrt{3} + 1 \notin Q$

$aR_2c \Rightarrow a^2 + c^2 = (2^{3/2})^2 + (1)^2$

$= 8 + 1 = 9 \in Q$

$\therefore R_2$ is not transitive.

33. (c) : R_2 is symmetric but R_2 is not transitive as

$(c, a), (a, b) \in R_2$ but $(c, b) \notin R_2$.

34. (d) : Here, $R_1 = \{(1, 8), (2, 6),$

$(3, 4), (4, 2)\}$ and $R_2 = \{(8, 1), (6, 2), (4, 3),$

$(2, 4)\}$

$\therefore$ Range of $R_2 = \{1, 2, 3, 4\}$

35. (c) : We have $(x, x) \in R$ for $w = 1$ implying that R is reflexive.

For $a \neq 0$, $(a, 0) \notin R$ for any w but $(0, a) \in R$. Thus R is not symmetric. Hence R is not an equivalence relation.

As $\left(\frac{m}{n}, \frac{m}{n}\right) \in S$

$\Rightarrow mn = mn$, S is reflexive.

$\left(\frac{m}{n}, \frac{p}{q}\right) \in S \Rightarrow qm = pn$

But this can be written as $np = mq$, giving

$\left(\frac{p}{q}, \frac{m}{n}\right) \in S$. Thus S is symmetric.

Again $\left(\frac{m}{n}, \frac{p}{q}\right) \in S$ and $\left(\frac{p}{q}, \frac{a}{b}\right) \in S$

means $qm = pn$ and $bp = aq$.

i.e., $\frac{m}{n} = \frac{p}{q}$ and $\frac{p}{q} = \frac{a}{b}$

i.e., $\frac{m}{n} = \frac{a}{b}$ Thus $\left(\frac{m}{n}, \frac{a}{b}\right) \in S$

This means S is transitive.

36. (a) : To be an equivalence relation the relation must be all reflexive, symmetric and transitive.

$T = \{(x, y) : x - y \in Z\}$ is reflexive - for $(x, x) \in Z$ i.e., $x - x = 0 \in Z$

symmetric - for $(x, y) \in Z \Rightarrow x - y \in Z$

$\Rightarrow y - x \in Z$ i.e. $(y, x) \in Z$

transitive - for $(x, y) \in Z$ and $(y, w) \in Z$

$\Rightarrow x - y \in Z$ and $y - w \in Z$,

giving $x - w \in Z$ i.e. $(x, w) \in Z$.

$\therefore T$ is an equivalence relation on R .

$S = \{(x, y) : y = x + 1, 0 < x < 2\}$ is not reflexive for $(x, x) \in S$ would imply

$x = x + 1 \Rightarrow 0 = 1$ (impossible).

Thus S is not an equivalence relation.

37. (b)

38. (d) : For $(3, 9) \in R, (9, 3) \notin R$

$\therefore$ Relation is not symmetric which means our choice (a) and (b) are out of court. We need to prove reflexivity and transitivity.

For reflexivity $a \in R, (a, a) \in R$ which is hold i.e. R is reflexive. Again, for transitivity of

$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$

which is also true in $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$.

39. (a) : R is not a function as

$A = \{1, 2, 3, 4\}$ and $(2, 4) \in R$ and $(2, 3) \in R$

R is not reflexive as $(1, 1) \notin R$

R is not symmetric as $(2, 3) \in R$ but $(3, 2) \notin R$

R is not transitive as $(1, 3) \in R$ and $(3, 1) \in R$ but $(1, 1) \notin R$.

40. (6) : Given, $A = \{1, 2, 3\}$

Let the relation be R on A , which is reflexive and transitive but not symmetric, then

$(1, 1), (2, 2), (3, 3), (1, 2) \in R$

Remaining elements are

$(2, 1), (2, 3), (1, 3), (3, 1), (3, 2)$

Case I : If relation contains exactly 4 elements $\Rightarrow 1$ way

Case II : If relation contains exactly 5 elements, so we can add $(1, 3)$ or $(3, 2) \Rightarrow 2$ ways

Case III : If relation contains exactly 6 elements, so we can add $(2, 3), (1, 3)$ or $(1, 3), (3, 2)$, or $(3, 1), (3, 2) \Rightarrow 3$ ways

$\therefore$ Total number of relations is 6.

41. (3) : For transitive : $(1, 2)$ and $(2, 3) \in R \Rightarrow (1, 3) \in R$

For reflexive : (1, 1), (2, 2), (3, 3) $\in R$

Now, for (2, 1), (3, 2), (3, 1);

(3, 1) cannot be taken for not symmetric relation.

Case-I : (2, 1) taken and (3, 2) not taken

Case-II : (3, 2) taken and (2, 1) not taken

Case-III : (2, 1) and (3, 2) are not taken

Therefore, 3 relations are possible.

42. (105) : 6 at of the 10 elements must be (a, a), (b, b), (f, f), for reflexive

For the remaining 4 spots, we can choose two pairs of form (a, b) and (b, a) $\in R$

Total Number of pairs to pick = ${}^6C_2 = 15$

Out of these number of ways to pick two pairs = ${}^{15}C_2 = 15 \times 7 = 105$

43. (960) : Let $A = \{1, 2, 3, 4\}$

$\therefore n(A) = 4$

$\therefore$ Number of reflexive and symmetric

relation = $2^{\frac{n^2-n}{2}} = 2^6$

Number of symmetric relation = $2^{n(n+1)/2} = 2^{10}$

$\therefore$ Number of relation which is symmetric but not reflexive = $2^{10} - 2^6 = 1024 - 64 = 960$

44. (16) : Given $A = \{1, 2, 3, 4\}$

$R = \{(1, 2), (2, 3), (1, 4)\}$

S to be equivalence, it should be reflexive, symmetric and transitive.

For reflexive add (1, 1) (2, 2), (3, 3), (4, 4).

For symmetric add (2, 1), (3, 2), (4, 1).

For transitive (1, 2), (2, 3) $\Rightarrow$ (1, 3), so add (1, 3) also add (3, 1) for symmetric and (4, 1), (1, 2) $\Rightarrow$ (4, 2), so add (4, 2) also add (2, 4) for symmetric.

$\therefore S = \{(1, 2), (2, 3), (1, 4), (1, 1), (2, 2), (3, 3), (4, 4), (2, 1), (3, 2), (4, 1), (3, 1), (1, 3), (2, 4), (4, 2), (3, 4), (4, 3)\}$

$\therefore n(S) = 16$

45. (66) : We have, $A = \{1, 2, 3, \dots, 100\}$

$R = \{(3, 2), (6, 4), (9, 6), (12, 8), \dots, (99, 66)\}$
 $n(R) = 33$

Since, R_1 is symmetric and $R \subset R_1$

$\therefore n(R_1) = 66$

46. (19) : $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$

$R = \{(x, y) \in A \times A : x - y \text{ is odd positive integer or } x - y = 2\}$

$\therefore$ Possible pairs to be added are $\{(0, 3), (0, 7), (0, 9), (3, 4), (3, 6), (3, 8), (3, 10), (4, 7), (4, 9), (6, 7), (6, 9), (7, 8), (7, 10), (8, 9), (9, 10), (4, 6), (6, 8), (7, 9), (8, 10)\}$

$\therefore$ We need to add minimum 19 elements to form it symmetric.

47. (3) : Let we have set $A = \{1, 2, 3\}$

By definition, we can say that

For reflexivity ; (1, 1) (2, 2) and (3, 3) $\in R$... (i)

For symmetry ; (2, 1) (1, 2) $\in R$... (ii)

For transitivity (1, 2) $\in R$ and (2, 3) $\in R$

$\Rightarrow (1, 3) \in R$... (iii)

But according to the question,

For not symmetric; (2, 1)

and (3, 2) $\notin R$... (iv)

So, $R_1 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$

$R_2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3), (2, 1)\}$ and $R_3 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3), (3, 2)\}$

48. (7) : $R = \{(-4, 4), (-3, 3), (3, -2), (0, 1), (0, 0), (1, 1), (4, 4), (3, 3)\}$

For reflexive, number of elements added to the relation R is $(-2, 2), (-4, -4), (-3, -3)$.

For Symmetric, number of elements added to the relation is $(4, -4), (3, -3), (-2, 3), (1, 0)$.

So, total number of elements = $3 + 4 = 7$

49. (13) : Given $S = \{(a, b), (b, c), (b, d)\}$

Reflexive (a, a), (b, b), (c, c), (d, d)

Symmetric (b, a), (c, b), (d, b)

Transitive (a, c), (a, d), (d, c)

$\downarrow \quad \downarrow \quad \downarrow$
(c, a) (d, a) (c, d)

So, 13 element must be added.

50. (b) : $R = \{(0, 3), (3, 0), (0, 4), (4, 0), (1, 3), (3, 1), (2, 3), (3, 2), (3, 3), (1, 4), (4, 1), (2, 4), (4, 2), (3, 4), (4, 3), (4, 4)\}$

$\therefore$ Number of elements in $R = 16$

$\therefore S_1$ is false.

Now, $(0, 0), (1, 1), (2, 2), (5, 5) \notin R \therefore R$ is not reflexive.

Again, let $(a, b) \in R$ then $(b, a) \in R$

As $\max\{a, b\} = \max\{b, a\} \therefore R$ is symmetric.

Now, R is not transitive as $(0, 3), (3, 1) \in R$ but $(0, 1) \notin R$.

$\therefore S_2$ is true.

51. (a) : Statement I :

Reflexive : $(a_1, b_1) R(a_1, b_1) \Rightarrow b_1 = b_1$

$\therefore R$ is reflexive.

Symmetric : $(a_1, b_1) R(a_2, b_2) \Rightarrow b_1 = b_2$

$(a_2, b_2) R(a_1, b_1) \Rightarrow b_2 = b_1$

$\therefore R$ is symmetric.

Transitive : $(a_1, b_1) R(a_2, b_2) \Rightarrow b_1 = b_2$

and $(a_2, b_2) R(a_3, b_3) \Rightarrow b_2 = b_3 \Rightarrow b_1 = b_3$
 $\Rightarrow (a_1, b_1) R(a_3, b_3)$

Hence, relation R is an equivalence relation.

$\therefore$ Statement I is true.

For statement II : $(x, y) R(a, b)$, for some $(a, b) \in X \Rightarrow y = b$

So, statement II is false.

52. (c) : $(x_1, y_1) R(x_2, y_2)$

$\Rightarrow x_1 \leq x_2$ or $y_1 \leq y_2$

For reflexive :

$(x_1, y_1) R(x_1, y_1)$

$\Rightarrow x_1 \leq x_1$ or $y_1 \leq y_1$ which is true.

So, R is reflexive.

For symmetric :

When $(x_1, y_1) R(x_2, y_2)$

$\Rightarrow x_1 \leq x_2$ or $y_1 \leq y_2$

$\nRightarrow x_2 \leq x_1$ or $y_2 \leq y_1$

They may or may not be true.

For example (1, 2) and (3, 4)

$1 \leq 3$ and $2 \leq 4$ but $3 \nless 1$ and $4 \nless 2$.

$\therefore R$ is not symmetric.

For transitive :

Take pairs as (3, 9), (4, 6), (2, 7)

(3, 9) R (4, 6) as $4 \geq 3$

(4, 6) R (2, 7) as $7 \geq 6$

But (3, 9) $\n R$ (2, 7), as neither $2 \geq 3$ nor $7 \geq 9$

So, R is not transitive.

53. (a) : $y - x = \text{integer}$

and $z - y = \text{integer} \Rightarrow z - x = \text{integer}$

$\therefore (x, y) \in A$ and $(y, z) \in A$

$\Rightarrow (x, z) \in A \Rightarrow$ Transitive

Also $(x, x) \in A$ is true $\Rightarrow$ Reflexive

As $(x, y) \in A \Rightarrow (y, x) \in A \Rightarrow$ Symmetric

Hence A is an equivalence relation but B is not.

Since, $(0, y)$ is in B but $(y, 0)$ is not in B .

54. We shall prove that R is reflexive, symmetric and transitive

Given $z_1 R z_2 \Leftrightarrow \frac{z_1 - z_2}{z_1 + z_2}$ is real.

$\therefore z_1 R z_1 \Leftrightarrow \frac{z_1 - z_1}{z_1 + z_1} = \frac{0}{2z_1} = 0$ purely real

$\therefore$ Relation is reflexive

Let $z_1 R z_2 \Leftrightarrow \frac{z_1 - z_2}{z_1 + z_2}$ is real

$\Leftrightarrow -\left(\frac{z_1 - z_2}{z_1 + z_2}\right)$ is real

$\Leftrightarrow \frac{z_2 - z_1}{z_2 + z_1}$ is real $\Leftrightarrow z_2 R z_1$

$\therefore$ Relation is symmetric

Let $z_1 R z_2$ and $z_2 R z_3$.

To show that $z_1 R z_3$.

Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2, z_3 = x_3 + iy_3$.

$\frac{z_1 - z_2}{z_1 + z_2}$ is real

$\Rightarrow \frac{(x_1 - x_2) + i(y_1 - y_2)}{(x_1 + x_2) + i(y_1 + y_2)}$ is real

$\frac{[(x_1 - x_2) + i(y_1 - y_2)][(x_1 + x_2) - i(y_1 + y_2)]}{(x_1 + x_2)^2 + (y_1 + y_2)^2}$ is real

$\Rightarrow (y_1 - y_2)(x_1 + x_2) - (x_1 - x_2)(y_1 + y_2) = 0$

$\Rightarrow 2x_2y_1 - 2y_2x_1 = 0 \Rightarrow \frac{x_1}{y_1} = \frac{x_2}{y_2}$... (1)

Similarly $z_2 R z_3 \Rightarrow \frac{x_2}{y_2} = \frac{x_3}{y_3}$

$\therefore \frac{x_1}{y_1} = \frac{x_3}{y_3} \Rightarrow z_1 R z_3$.

$\therefore$ Relation is Transitive

$\therefore$ It is an equivalence relation.

55. (d) : We have,

$$f(x) = \log_e \left(\frac{2x-3}{5+4x} \right) + \sin^{-1} \left(\frac{4+3x}{2-x} \right)$$

For $f(x)$ to be defined we have,

$$\frac{2x-3}{5+4x} > 0 \text{ and } \left| \frac{4+3x}{2-x} \right| \leq 1$$

$$\text{Now, } \frac{2x-3}{5+4x} > 0$$

Case-I : $2x-3 > 0$ and $5+4x > 0$

$$\Rightarrow x > 3/2 \text{ and } x > -5/4$$

$$\Rightarrow x \in (3/2, \infty) \quad \dots(i)$$

Case-II : $2x-3 < 0$ and $5+4x < 0$

$$\Rightarrow x < 3/2 \text{ and } x < -5/4$$

$$\Rightarrow x \in (-\infty, -5/4) \quad \dots(ii)$$

From (i) and (ii), we get

$$x \in \left(-\infty, -\frac{5}{4} \right) \cup \left(\frac{3}{2}, \infty \right) \quad \dots(iii)$$

$$\text{Also, } \left| \frac{4+3x}{2-x} \right| \leq 1$$

$$\Rightarrow -1 \leq \frac{4+3x}{2-x} \leq 1$$

$$\Rightarrow -1 \leq \frac{4+3x}{2-x} \text{ and } \frac{4+3x}{2-x} \leq 1$$

$$\Rightarrow 0 \leq \frac{4+3x}{2-x} + 1 \text{ and } \frac{4+3x}{2-x} - 1 \leq 0$$

$$\Rightarrow 0 \leq \frac{6+2x}{2-x} \text{ and } \frac{2+4x}{2-x} \leq 0$$

$$\Rightarrow -3 \leq x \text{ and } x \leq -\frac{1}{2}, \quad x \neq 2$$

$$\Rightarrow x \in \left[-3, -\frac{1}{2} \right] \quad \dots(iv)$$

From (iii) and (iv), we get

$$x \in \left[-3, -\frac{5}{4} \right]$$

$$\therefore \alpha = -3 \text{ and } \beta = -\frac{5}{4}$$

$$\text{Thus, } \alpha^2 + 4\beta = 9 - 5 = 4$$

56. (a) : Given $f, g : (1, \infty) \rightarrow R, f(x) :$

$$\frac{2x+3}{5x+2}, g(x) = \frac{2-3x}{1-x}$$

also, we have $f \circ g : [2, 4] \rightarrow R$

$$\text{Now, } g(2) = \frac{2-6}{1-2} = 4, g(4) = \frac{2-12}{1-4} = \frac{10}{3}$$

$$\therefore f(g(2)) = \frac{8+3}{20+2} = \frac{1}{2}, f(g(4)) = \frac{20+9}{50+6} = \frac{29}{56}$$

$$\text{i.e., } \alpha = \frac{1}{2} \text{ and } \beta = \frac{29}{56}$$

$$\text{Then, } \frac{1}{\beta - \alpha} = \frac{1}{\frac{29}{56} - \frac{1}{2}} = \frac{1}{\frac{1}{56}} = 56$$

57. (c) : We have $A : x^2 + y^2 = 25 \quad \dots(i)$

$$B : \frac{x^2}{144} + \frac{y^2}{16} = 1 \quad \dots(ii)$$

$$C : x^2 + y^2 \leq 4 \quad \dots(iii)$$

Solving (i) and (ii), we get

$$x^2 + 9(25 - x^2) = 144$$

$$\Rightarrow -8x^2 = -81 \Rightarrow x = \pm \frac{9}{2\sqrt{2}}$$

From (i),

$$\frac{81}{8} + y^2 = 25 \Rightarrow y^2 = 25 - \frac{81}{8} \Rightarrow y = \pm \frac{\sqrt{119}}{2\sqrt{2}}$$

As, $D = A \cap B$

$$= \left\{ \left(\frac{9}{2\sqrt{2}}, \frac{\sqrt{119}}{2\sqrt{2}} \right), \left(\frac{9}{2\sqrt{2}}, -\frac{\sqrt{119}}{2\sqrt{2}} \right), \left(-\frac{9}{2\sqrt{2}}, \frac{\sqrt{119}}{2\sqrt{2}} \right), \left(-\frac{9}{2\sqrt{2}}, -\frac{\sqrt{119}}{2\sqrt{2}} \right) \right\}$$

$$\Rightarrow n(D) = 4$$

Also $C = \{x, y\} \in Z \times Z : x^2 + y^2 \leq 4\}$

$$= \{(0, 2), (0, -2), (2, 0), (-2, 0), (1, 1), (-1, -1), (-1, 1), (1, -1), (0, 1), (0, -1), (1, 0), (-1, 0), (0, 0)\}$$

$$\Rightarrow n(C) = 13.$$

$\therefore$ Total number of one - one function from D to $C = {}^{13}P_4 = 17160$

58. (c) : Here, $n(A) = 4, n(B) = 4$

Total number of functions from A to $B = 4^4 = 256$

Number of one-one functions from A to $B = {}^4P_4 = 4! = 24$

Number of many-one functions from A to $B = 256 - 24 = 232$

Number of many-one function for which $1 \notin f(A) = 3^4 = 81$

$\therefore$ Required number of many-one functions

$$= 232 - 81 = 151$$

59. (d) : Given,

$$g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$$

Here, $Dg \in R \quad [\because 2x^2 - 2x + 1 > 0]$

Also, $f(x) = \log_e x \Rightarrow D_f \in (0, \infty)$

$$\therefore D_{f \circ g} \Rightarrow g(x) > 0$$

$$\Rightarrow \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1} > 0$$

$$\Rightarrow x^4 - 2x^3 + 3x^2 - 2x + 2 > 0$$

The above expression is always positive for any value of x .

$\therefore$ Domain of $f \circ g \in R$.

60. (c) : Given function is

$$f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}; f(x) = \frac{2^x - \frac{1}{2^x}}{2^x + \frac{1}{2^x}}$$

$$= \frac{2^{2x} - 1}{2^{2x} + 1} = 1 - \frac{2}{2^{2x} + 1} \Rightarrow f'(x)$$

$$= \frac{2}{(2^{2x} + 1)^2} \times 2 \times 2^{2x} \times \log_e(2) > 0$$

$\Rightarrow f(x)$ is always increasing, Hence it is one-one.

Since, $f(-\infty) = -1$ and $f(\infty) = 1$

$$\Rightarrow f(x) \in (-1, 1) \neq (-\infty, \infty)$$

Thus, the function $f(x)$ is one-one but not onto.

61. (c) : As $f(x)$ is onto, hence A is range of $f(x)$.

Now $f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3)$

For extremum, $f(2) = 16 - 60 + 72 + 7 = 35$;

$f(3) = 54 - 135 + 108 + 7 = 34$; $f(0) = 7$; $F(1) = 30$.

Hence, range $\in [7, 35] = A$

Also, for range of $g(x)$,

$$g(x) = 1 - \frac{1}{x^{2025} + 1} \in [0, 1) = B$$

$$\therefore S = \{0, 7, 8, \dots, 35\}$$

Hence, $n(S) = 30$

62. (b) : We have, $(g \circ f)(x) = x$

$$\Rightarrow g(f(x)) = x$$

Differentiating w.r.t. x , we get

$$g'(f(x)) \cdot f'(x) = 1$$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

Now, $f(x) = x^5 + 2e^{x/4}$

$$\Rightarrow f'(x) = 5x^4 + \frac{1}{2}e^{x/4}$$

$$\therefore f'(0) = \frac{1}{2}, f(0) = 2 \Rightarrow g'(2) = 2$$

Hence, $8g'(2) = 16$

63. (d) : $A = \{1, 3, 7, 9, 11\}, B = \{2, 4, 5, 7, 8, 10, 12\}$

$f : A \rightarrow B$ is one-one such that

$$f(1) + f(3) = 14$$

$$(f(1), f(3)) = \{(2, 12), (4, 10), (12, 2), (10, 4)\}$$

Since, f is one-one, so $f(1) \neq f(3)$, so we can not take $(7, 7)$.

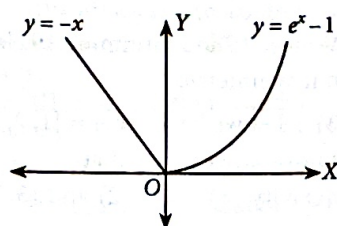
$\therefore$ So, for $f(1)$, we have 4 choices and for $f(3)$, we have 4 choices and remaining 3 elements have 5! choices for mapping to be one-one.

Total number of ways = $4 \times 5! / 2$

$$= \frac{480}{2} = 240 \quad [\because \text{Pairs } (2, 12) \text{ and } (12, 2) \text{ will be considered same}]$$

64. (b) : $f(g(x)) = |g(x) - 1|$

$$= \begin{cases} |e^x - 1|, & x \geq 0 \\ |x + 1 - 1|, & x \leq 0 \end{cases} = \begin{cases} e^x - 1, & x \geq 0 \\ -x, & x \leq 0 \end{cases}$$



$f(g(x))$ is neither one-one nor onto as negative numbers have no pre-image.

65. (b) : We have, $f(x) = \frac{x^2 + 2x - 15}{x^2 - 4x + 9}$

$$f(x) = \frac{(x+5)(x-3)}{x^2 - 4x + 9}$$

$f(x)$ has two roots as $x = -5, 3$

So $f(-5) = 0$ and $f(3) = 0$

So f can not be one-one.

Now consider, $x^2 - 4x + 9$

$$D = 16 - 36 = -20 < 0$$

Now, Let $y = \frac{x^2 + 2x - 15}{x^2 - 4x + 9}$

$$\Rightarrow yx^2 - 4xy + 9y = x^2 + 2x - 15$$

$$\Rightarrow x^2(y-1) - 2x(2y+1) + (9y+15) = 0$$

$$D = 4(2y+1)^2 - 4(y-1)(9y+15) \geq 0$$

[$\because x \in R$]

$$\Rightarrow 4(4y^2 + 1 + 4y) - (36y^2 - 36y + 60y - 60) \geq 0$$

$$\Rightarrow 16y^2 + 4 + 16y - 36y^2 - 24y + 60 \geq 0$$

$$\Rightarrow -20y^2 - 8y + 64 \geq 0$$

$$\Rightarrow 5y^2 + 2y - 16 \leq 0$$

$$\Rightarrow (5y-8)(y+2) \leq 0$$

$$\Rightarrow y \in \left[-2, \frac{8}{5}\right] \text{ which is range of function } f$$

So, f is not onto if $f: R \rightarrow R$.

Here in given question co-domain is not defined.

66. (d) : $2310 = 2 \times 3 \times 5 \times 7 \times 11$

$$\therefore A = \{2, 3, 5, 7, 11\}$$

$f: A \rightarrow Z$ is a function such that

$$f(x) = \left\lfloor \log_2 \left(x^2 + \left\lfloor \frac{x^3}{5} \right\rfloor \right) \right\rfloor$$

$$f(2) = \lfloor \log_2(4 + \lfloor 1.6 \rfloor) \rfloor = \lfloor \log_2(4 + 1) \rfloor = \lfloor \log_2 5 \rfloor$$

$$= \lfloor \log_2(2^2 + 1) \rfloor = 2$$

$$f(3) = \lfloor \log_2(9 + 5) \rfloor = \lfloor \log_2(2^3 + 6) \rfloor = 3$$

$$f(5) = \lfloor \log_2(25 + 25) \rfloor = \lfloor \log_2(2^5 + 18) \rfloor = 5$$

$$f(7) = \lfloor \log_2(49 + 68) \rfloor = \lfloor \log_2(2^6 + 53) \rfloor = 6$$

$$f(11) = \lfloor \log_2(121 + 266) \rfloor = \lfloor \log_2(2^8 + 131) \rfloor$$

$$= 8$$

$$\text{Range of } f = \{2, 3, 5, 6, 8\}$$

$$\text{Number of one-one functions} = 5! = 120$$

67. (d) : $y = f(x) = \begin{cases} -a, & -a \leq x \leq 0 \\ x+a, & 0 < x \leq a \end{cases}$

$$y = f|x| = \begin{cases} -a, & -a \leq |x| \leq 0 \\ |x|+a, & 0 < |x| \leq a \end{cases}$$

but $|x| < 0$ is not possible.

$$f|x| = \begin{cases} -x+a, & -a \leq x < 0 \\ x+a, & 0 < x \leq a \end{cases}$$

$$y = |f(x)| = \begin{cases} a, & -a \leq x \leq 0 \\ x+a, & 0 < x \leq a \end{cases}$$

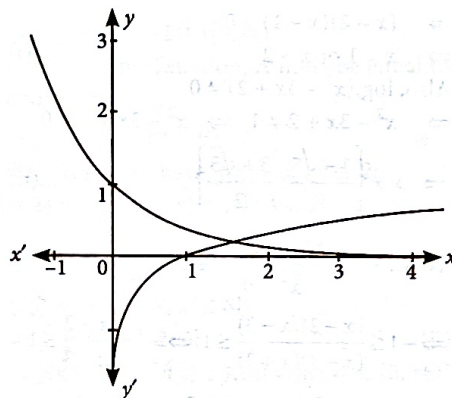
$$g(x) = \frac{f|x| - |f(x)|}{2}$$

$$= \begin{cases} \frac{-x+a-a}{2} = -\frac{x}{2} & \text{if } -a \leq x \leq 0 \\ \frac{x+a-x-a}{2} = 0 & \text{if } 0 < x \leq a \end{cases}$$

$g: [-a, a] \rightarrow [-a, a]$ is neither one-one nor onto as set $[0, a]$ has only one image i.e. 0.

68. (a) : $g(f(x)) = \begin{cases} f(x), & f(x) \geq 0 \\ e^{f(x)}, & f(x) < 0 \end{cases}$

$$= \begin{cases} \log_e x, & x \geq 1 \\ e^{-x}, & x \leq 0 \\ e^{\log_e x} = x, & 0 < x < 1 \end{cases}$$



$$gof(x) = \begin{cases} e^{-x}, & x \leq 0 \\ x, & 0 < x < 1 \\ \log_e x, & x \geq 1 \end{cases}$$

Now, range of $gof = [0, \infty) \neq R$

Hence, not one-one and not onto.

69. (d) : There are two cases arise :

Case I : Let a is even $\therefore f(a) = a - 1$ (odd)

$$\Rightarrow f(f(a)) = f(a-1) = 2a-2 \text{ (even)}$$

$$\Rightarrow f(f(f(a))) = f(2a-2)$$

$$= 2a-2-1 = 2a-3$$

$$\Rightarrow 21 = 2a-3$$

$$\Rightarrow a = 12$$

So,

$$\lim_{x \rightarrow a^-} \left\{ \frac{|x|^3}{a} - \left\lfloor \frac{x}{a} \right\rfloor \right\} = \lim_{x \rightarrow 12^-} \left\{ \frac{|x|^3}{12} - \left\lfloor \frac{x}{12} \right\rfloor \right\}$$

$$= 144$$

($\because x < 12$)

Case II : Let a is odd

$$\therefore f(a) = 2a \text{ (even)}$$

$$\Rightarrow f(f(a)) = f(2a) = 2a-1 \text{ (odd)}$$

$$\Rightarrow f(f(f(a))) = f(2a-1) = 2(2a-1)$$

$$\Rightarrow 21 = 4a-2 \Rightarrow a = \frac{23}{4} \notin N$$

70. (c) : Given, $f: N - \{1\} \rightarrow N$

$f(n)$ = the highest prime factor of n

For one-one : If $n = 4, f(n) = 2$.

If $n = 8, f(n) = 2 \therefore f$ is not one-one.

For onto : Range = All prime numbers

Co-domain = Set of natural numbers

$\Rightarrow$ Range $\neq$ Co-domain $\therefore f$ is not onto.

71. (d)

72. (d) : Given,

$$f(x) = \begin{cases} 2+2x, & -1 \leq x < 0 \\ 1-\frac{x}{3}, & 0 \leq x \leq 3 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} -x, & -3 \leq x \leq 0 \\ x, & 0 < x \leq 1 \end{cases}$$

$$fog(-3) = f(3) = 0; fog(-2) = f(2) = \frac{1}{3};$$

$$fog(-1) = f(1) = \frac{2}{3}$$

$$fog(0) = f(0) = 1; fog(1) = f(1) = \frac{2}{3}$$

$$\therefore \text{Range of } fog(x) = [0, 1]$$

73. (d) : Here, $f(x) = \frac{4x+3}{6x-4}$

$$f[f(x)] = \frac{4f(x)+3}{6f(x)-4} = \frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}$$

$$= \frac{16x+12+18x-12}{6x-4} = \frac{34x}{6x-4} = x$$

$$\therefore f[f(x)] = x \Rightarrow g(x) = x$$

$$\text{Now, } (gogog)(4) = gog[g(4)] = gog(4) = g(4) = 4$$

74. (a) : Given, $f(x) = e^{x^3-3x+1}$

$$\therefore f'(x) = e^{x^3-3x+1} [3(x-1)(x+1)] \geq 0$$

As, e^{x^3-3x+1} is always positive.

$$\therefore 3(x-1)(x+1) \geq 0$$



$$\Rightarrow x \in (-\infty, -1] \cup [1, \infty)$$

For onto function, Range = Co-domain

$$\therefore a = f(-\infty) = e^{-\infty} = 0$$

$$\text{and } b = f(-1) = e^{-1+3+1} = e^3$$

$$\therefore P(2b+4, a+2) = P(2e^3+4, 2)$$

Now, distance of point $P(2e^3+4, 2)$ from the line $x + e^3y - 4 = 0$

$$= \frac{2e^3+4+2e^{-3}-4}{\sqrt{1+e^{-6}}}$$

$$= \frac{2(e^3+e^{-3})}{\sqrt{1+e^{-6}}} = 2\left(\frac{e^6+1}{e^3}\right) \times \frac{1}{\sqrt{1+e^6}} \times e^3$$

$$= 2\sqrt{1+e^6}$$

75. (a) : Let $f(x) = \frac{Ax+B}{Cx-A}$

$$f(f(x)) = \frac{A\left(\frac{Ax+B}{Cx-A}\right)+B}{C\left(\frac{Ax+B}{Cx-A}\right)-A} = \frac{A^2x+BCx}{BC+A^2} = x \quad \dots(i)$$

$$f\left(f\left(\frac{4}{x}\right)\right) = \frac{4}{x} \quad [\text{From (i)}]$$

$$\therefore f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right) = x + \frac{4}{x} \geq 4$$

[Using A.M. $\geq$ G.M.]

76. (*) : For $x \in \mathbb{R}$, $f(x)$ and $g(x)$ are real valued functions

$$g(x) = \sqrt{x} + 1 \text{ and } fog(x) = x + 3 - \sqrt{x}$$

$$fog(x) = x + 3 - \sqrt{x} \Rightarrow f(g(x)) = x + 3 - \sqrt{x}$$

$$= (\sqrt{x} + 1)^2 - 3(\sqrt{x} + 1) + 5$$

$$= [g(x)]^2 - 3[g(x)] + 5 \quad (\because g(x) = \sqrt{x} + 1)$$

$$\text{So, } f(x) = x^2 - 3x + 5$$

$$\therefore f(0) = 0 - 3 \times 0 + 5 = 5$$

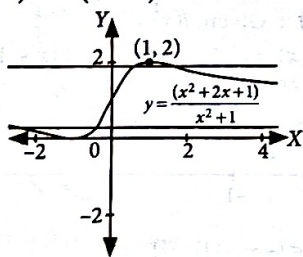
Note: But if we consider the domain of the composite function $fog(x)$, then $f(0)$ will not be defined as $g(x)$ can't be equal to zero.

77. (d) : Given, $f(x) = \frac{x^2 + 2x + 1}{x^2 + 1}$.

$$\text{So, } f(x) = 1 + \frac{2x}{x^2 + 1}$$

$$f'(x) = \frac{(x^2 + 1) \cdot 2 - 2x \times 2x}{(x^2 + 1)^2}$$

$$= \frac{2 - 2x^2}{(x^2 + 1)^2} = \frac{2(1 - x^2)}{(x^2 + 1)^2}$$



The graph of function is, By horizontal line test we can say that $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$.

78. (d)

79. (c) : $f(x) = \sin^{-1}[2x^2 - 3] +$

$$\log_2(\log_{1/2}(x^2 - 5x + 5))$$

$$\text{As, } -1 \leq [2x^2 - 3] \leq 1$$

$$\Rightarrow -1 \leq 2x^2 - 3 < 2$$

$$\Rightarrow -1 + 3 \leq 2x^2 < 2 + 3$$

$$\Rightarrow 2 \leq 2x^2 < 5$$

$$\Rightarrow 1 \leq x^2 < 5/2$$

$$\Rightarrow x \in \left[-\sqrt{\frac{5}{2}}, -1\right] \cup \left[1, \sqrt{\frac{5}{2}}\right] \quad \dots(1)$$

$$\text{Now, consider } \log_2(\log_{1/2}(x^2 - 5x + 5))$$

$$\log_{1/2}(x^2 - 5x + 5) > 0$$

$$\Rightarrow 0 < x^2 - 5x + 5 < 1$$

$$\Rightarrow x^2 - 5x + 5 > 0 \text{ and } x^2 - 5x + 4 < 0$$

$$\Rightarrow x \in \left(-\infty, \frac{5 - \sqrt{5}}{2}\right) \cup \left(\frac{5 + \sqrt{5}}{2}, \infty\right)$$

and $x \in (1, 4)$

So this implies that

$$x \in \left(1, \frac{5 - \sqrt{5}}{2}\right) \cup \left(\frac{5 + \sqrt{5}}{2}, 4\right) \quad \dots(2)$$

So, from (1) and (2), we get

$$x \in \left(1, \frac{5 - \sqrt{5}}{2}\right)$$

80. (*) : We have,

$$f(x) = \frac{\cos^{-1}\left(\frac{x^2 - 5x + 6}{x^2 - 9}\right)}{\log_e(x^2 - 3x + 2)}$$

For domain, $x^2 - 3x + 2 > 0$

$$\Rightarrow (x - 2)(x - 1) > 0$$

$$\Rightarrow x < 1 \text{ or } x > 2 \quad \dots(i)$$

Also, $\log_e(x^2 - 3x + 2) \neq 0$

$$\Rightarrow x^2 - 3x + 2 \neq 1 \Rightarrow x^2 - 3x + 1 \neq 0$$

$$\Rightarrow x \neq \left\{\frac{3 - \sqrt{5}}{2}, \frac{3 + \sqrt{5}}{2}\right\} \quad \dots(ii)$$

$$\text{Now, } -1 \leq \frac{x^2 - 5x + 6}{x^2 - 9} \leq 1$$

$$\Rightarrow -1 \leq \frac{(x - 2)(x - 3)}{(x - 3)(x + 3)} \leq 1 \Rightarrow -1 \leq \frac{x - 2}{x + 3} \leq 1$$

$$\text{Case I: } \frac{x - 2}{x + 3} + 1 \geq 0 \Rightarrow \frac{2x + 1}{x + 3} \geq 0$$

$$\Rightarrow x \geq -\frac{1}{2} \text{ or } x < -3 \quad \dots(iii)$$

$$\text{Case II: } \frac{x - 2}{x + 3} - 1 \leq 0 \Rightarrow \frac{-5}{x + 3} \leq 0$$

$$\Rightarrow x + 3 > 0 \Rightarrow x > -3 \quad \dots(iv)$$

$$\text{And, } x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3 \quad \dots(v)$$

So, required domain is,

$$\left[-\frac{1}{2}, 1\right) \cup (2, 3) \cup (3, \infty) - \left\{\frac{3 + \sqrt{5}}{2}, \frac{3 - \sqrt{5}}{2}\right\}$$

81. (b)

82. (d) : We have, $f(x) = x - 1 \quad x \in \mathbb{R}$

$$\text{and } g(x) = \frac{x^2}{x^2 - 1}, x \in \mathbb{R} - \{-1, 1\}$$

$$\therefore fog(x) = \frac{x^2}{x^2 - 1} - 1 = \frac{1}{x^2 - 1}$$

So, domain of $fog(x) = \mathbb{R} - \{-1, 1\}$

and range of $fog(x) = \mathbb{R} - (-1, 0)$

For one-one : Let $f(x_1) = f(x_2)$

$$\Rightarrow \frac{1}{x_1^2 - 1} = \frac{1}{x_2^2 - 1} \Rightarrow x_2^2 = x_1^2$$

$$\Rightarrow x_2 = \pm x_1 \Rightarrow fog(x) \text{ is not one-one.}$$

$$\text{Now, let } y = \frac{1}{x^2 - 1}$$

$$\Rightarrow x^2 - 1 = \frac{1}{y} \Rightarrow x^2 = \frac{1 + y}{y}$$

$$\Rightarrow x = \sqrt{\frac{1 + y}{y}}$$

For every $x \in \{-1, 1\}$ $fog(x)$ is not defined

$\therefore fog(x)$ is not onto.

83. (d) 84. (c) 85. (c)

86. (a) 87. (a)

88. (c) : We have, $f(x) = \frac{x - 2}{x - 3}$

$$\Rightarrow f^{-1}(x) = \frac{3x - 2}{x - 1}$$

$$\text{Also, } g(x) = 2x - 3 \Rightarrow g^{-1}(x) = \frac{x + 3}{2}$$

$$\text{Now, } f^{-1}(x) + g^{-1}(x) = \frac{13}{2} \text{ (given)}$$

$$\Rightarrow \left(\frac{3x - 2}{x - 1}\right) + \frac{(x + 3)}{2} = \frac{13}{2}$$

$$\Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 3, 2$$

$$\Rightarrow \text{Sum of all values of } x = 5$$

89. (c) : $f(g(x)) = 2g(x) - 1$

$$= 2\left(\frac{2x - 1}{2(x - 1)}\right) - 1 = \frac{x}{x - 1} = 1 + \frac{1}{x - 1}$$

Range of $f(g(x)) = \mathbb{R} - \{1\}$

Thus, range of $f(g(x))$ is not co-domain of $f(g(x))$.

$\therefore f(g(x))$ is not onto.

But $f(g(x))$ is one-one.

$$[\nless f(g(x_1)) = f(g(x_2)) \Rightarrow x_1 = x_2]$$

So, $f(g(x))$ is one-one but not onto.

90. (d)

91. (b) : $x = {}^5P_3 = 60$

$$y = {}^{15}P_3 = 15 \times 14 \times 13 = 30 \times 91$$

$$\therefore 2y = 91x$$

92. (d) : We have,

$$f(x) = \begin{cases} k + 1, & \text{if } k \text{ is odd} \\ k, & \text{if } k \text{ is even} \end{cases}$$

$\therefore g : A \rightarrow A$ such that $g(f(k)) = f(k)$

If k is even, then $g(k) = k \quad \dots(i)$

If k is odd, then $g(k + 1) = k + 1 \quad \dots(ii)$

From (i) and (ii) we can say that

$g(k) = k$ if k is even and if k is odd, then $g(k)$

can take any value in set A .

So, number of possible functions from A to

$$A = 10^5 \times 1 = 10^5$$

93. (d)

94. (b) : Since, $g(x) = x^2 + x - 1$

$$\therefore (gof)(x) = g(f(x)) = f^2(x) + f(x) - 1$$

$$\Rightarrow g\left(f\left(\frac{5}{4}\right)\right) = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1 \quad \dots(i)$$

$$\text{Again, } g\left(f\left(\frac{5}{4}\right)\right) = 4\left(\frac{5}{4}\right)^2 - 10\left(\frac{5}{4}\right) + 5 = -\frac{5}{4} \quad \dots(ii)$$

From (i) and (ii), we get

$$-\frac{5}{4} = f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) - 1$$

$$\Rightarrow f^2\left(\frac{5}{4}\right) + f\left(\frac{5}{4}\right) + \frac{1}{4} = 0$$

$$\Rightarrow \left(f\left(\frac{5}{4}\right) + \frac{1}{2}\right)^2 = 0 \Rightarrow f\left(\frac{5}{4}\right) = -\frac{1}{2}$$

$$95. (b) : \text{Let } f(x) = y = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$$

$$\Rightarrow \frac{1+y}{1-y} = \frac{8^{2x}}{8^{-2x}} \Rightarrow 8^{4x} = \frac{1+y}{1-y}$$

$$\Rightarrow 4x = \log_8 \left(\frac{1+y}{1-y} \right) \Rightarrow x = \frac{1}{4} \log_8 \left(\frac{1+y}{1-y} \right)$$

$$\Rightarrow f^{-1}(x) = \frac{1}{4} \log_8 \left(\frac{1+x}{1-x} \right) = \frac{1}{4} \frac{\log_e \left(\frac{1+x}{1-x} \right)}{\log_e 8}$$

$$= \frac{1}{4} (\log_8 e) \log_e \left(\frac{1+x}{1-x} \right)$$

$$96. (d) : f(x) = \begin{cases} \frac{x}{x^2+1}; & x \in (1, 2) \\ \frac{2x}{x^2+1}; & x \in [2, 3] \end{cases}$$

$\Rightarrow f(x)$ is a strictly decreasing function.

$$\therefore f(x) \in \left(\frac{2}{5}, \frac{1}{2} \right) \cup \left(\frac{6}{10}, \frac{4}{5} \right]$$

$$\Rightarrow f(x) \in \left(\frac{2}{5}, \frac{1}{2} \right) \cup \left(\frac{3}{5}, \frac{4}{5} \right]$$

$$97. (c) : f(x) = |x| (x - \sin x)$$

$$f(x) = \begin{cases} x^2 - x \sin x, & x \geq 0 \\ -x^2 + x \sin x, & x < 0 \end{cases}$$

f is continuous as it is a product of two continuous functions.

$$f(-\infty) = \lim_{x \rightarrow -\infty} (-x^2 + x \sin x)$$

$$= \lim_{x \rightarrow -\infty} (-x^2) \left(1 - \frac{\sin x}{x} \right) = -\infty$$

$$f(\infty) = \lim_{x \rightarrow \infty} (x^2 - x \sin x)$$

$$= \lim_{x \rightarrow \infty} (x^2) \left(1 - \frac{\sin x}{x} \right) = \infty$$

So, the range of f is R .

Then f is ONTO.

Again,

$$f'(x) = \begin{cases} 2x - x \cos x - \sin x, & x > 0 \\ -2x + \sin x + x \cos x, & x < 0 \end{cases}$$

Note that, $2x - x \cos x - \sin x$

$$= \underbrace{(x - \sin x)}_{\geq 0} + \underbrace{x(1 - \cos x)}_{\geq 0}$$

Thus, $f' \geq 0 \forall x \in (0, \infty)$

Similarly, $f' \geq 0 \forall x \in (-\infty, 0)$ as f' is even because f is odd.

Thus, f is ONE-ONE.

$$98. (d) : \text{Given, } f(x) = \frac{x^2}{1-x^2} = y (\text{say})$$

$$\Rightarrow x^2 = y - x^2 y \Rightarrow (1+y)x^2 = y$$

$$\Rightarrow x^2 = \frac{y}{1+y} \geq 0$$

$$\Rightarrow y \in (-\infty, -1) \cup [0, \infty)$$

For surjective function, A must be equal to above range.

$$\therefore A = R - [-1, 0)$$

$$99. (a) : \text{We have, } f(x) = x^2, x \in R;$$

$$g(A) = \{x \in R : f(x) \in A\}, A \subseteq R$$

$$\text{and } S = [0, 4]$$

$$\text{Now, } f(S) = S^2 \Rightarrow f(S) \in [0, 16] \quad \dots(i)$$

$$g(S) = \{x : x \in R, x^2 \in S\}$$

$$\Rightarrow g(S) \in [-2, 2] \quad \dots(ii)$$

$$\therefore g(f(S)) = \{x : f(x) \in f(S)\} = \{x : x^2 \in [0, 16]\}$$

$$\Rightarrow g(f(S)) \in [-4, 4] \quad \dots(iii)$$

$$\text{From (ii) and (iii), } g(f(S)) \neq g(S)$$

$$g(f(S)) \neq S \text{ Now, } f(g(S)) = (g(S))^2$$

$$\Rightarrow f(g(S)) \in [0, 4] \quad \dots(iv)$$

$$\therefore f(g(S)) = S$$

$$\text{From (i) and (iv), } f(g(S)) \neq f(S)$$

$$100. (d) : \text{Here, } f(x) = \log_e(\sin x),$$

$$0 < x < \pi \text{ and } g(x) = \sin^{-1}(e^{-x}), x \geq 0$$

$$\text{Now, } (f \circ g)(x) = \log_e(\sin(\sin^{-1}(e^{-x})))$$

$$= \log(e^{-x}) = -x$$

$$\therefore (f \circ g)'(x) = -1$$

$$\text{Now, } (f \circ g)(\alpha) = -\alpha = b$$

$$\text{and } (f \circ g)'(\alpha) = -1 = a \quad (\text{Given})$$

$$\therefore (a) \alpha a^2 + b\alpha + a = -b^2 + b(-b) - 1$$

$$= -b^2 - b^2 - 1 = -2b^2 - 1 \neq 0$$

$$(b) \alpha a^2 - b\alpha - a = -b^2 - b(-b) - (-1)$$

$$= -b^2 + b^2 + 1 = 1 \neq 0$$

$$(c) \alpha a^2 + b\alpha - a = -b^2 + b(-b) - (-1)$$

$$= -b^2 - b^2 + 1 = -2b^2 + 1 = -2\alpha^2 + 1 \neq -2\alpha^2$$

$$(d) \alpha a^2 - b\alpha - a = -b^2 - b(-b) - (-1)$$

$$= -b^2 + b^2 + 1 = 1$$

Hence, only option (d) satisfies.

$$101. (d) : \text{We have,}$$

$$f(x) = \sqrt{x}, g(x) = \tan x \text{ and } h(x) = \frac{1-x^2}{1+x^2}$$

$$\therefore f \circ g(x) = f(g(x)) = \sqrt{\tan x}$$

$$\text{Now, } \phi(x) = ((h \circ f) \circ g)(x) = h(f(g(x)))$$

$$= \frac{1 - \tan x}{1 + \tan x} = \tan\left(\frac{\pi}{4} - x\right)$$

$$\therefore \phi(x) = \tan\left(\frac{\pi}{4} - x\right)$$

$$\text{At } x = \frac{\pi}{3}, \phi\left(\frac{\pi}{3}\right) = \tan\left(\frac{\pi}{4} - \frac{\pi}{3}\right)$$

$$= \tan\left(\frac{-\pi}{12}\right) = \tan\frac{11\pi}{12}$$

102. (c)

$$103. (b) : \text{We have, } f(x) = \frac{2x}{x-1}$$

Since x is not a +ve integer.

$\therefore$ For, $x = 0, f(x) = 0$; For, $x < 0,$

$$f(x) \in (0, 2)$$

So, clearly $f(x)$ is not onto as its range only belongs to $(0, 2)$ not to R .

$$\text{Now, } f'(x) = \frac{-2}{(x-1)^2}, \text{ which is always less than zero.}$$

$$\Rightarrow f(x) \text{ is decreasing}$$

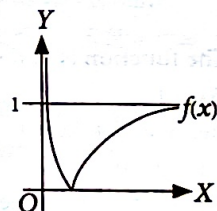
$$\therefore f(x) \text{ is one-one.}$$

Hence, $f(x)$ is injective but not surjective.

104. (d)

105. (*) : Here,

$$f(x) = \left| 1 - \frac{1}{x} \right| = \begin{cases} \frac{1-x}{x}, & 0 < x < 1 \\ \frac{x-1}{x}, & x \geq 1 \end{cases}$$



$\therefore f(x)$ is not injective but range of function is $[0, \infty)$.

Note: If co-domain is $[0, \infty)$, then $f(x)$ will be surjective.

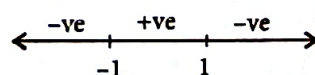
106. (a)

$$107. (b) : \text{We have, } f(x) = \frac{x}{1+x^2}$$

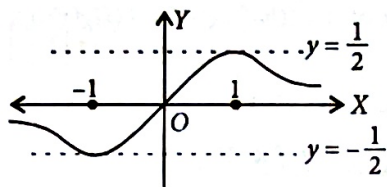
$$\therefore f'(x) = \frac{(1+x^2) \cdot 1 - x \cdot 2x}{(1+x^2)^2}$$

$$= \frac{1-x^2}{(1+x^2)^2} = \frac{(1-x)(1+x)}{(1+x^2)^2}$$

The sign of $f'(x)$ is given as



Now, f can be graphed as under



Clearly function is surjective but not injective, as a horizontal line meet the curve in two points.

108. (d) : We have, $f(g(x)) = x$

$$f(3^{10}x - 1) = 2^{10}(3^{10}x - 1) + 1 = x$$

$$\Rightarrow 2^{10} \cdot 3^{10}x - 2^{10} + 1 = x$$

$$\Rightarrow x(2^{10}3^{10} - 1) = 2^{10} - 1$$

$$\Rightarrow x = \frac{2^{10} - 1}{2^{10}3^{10} - 1} \Rightarrow x = \frac{2^{10}(1 - 2^{-10})}{2^{10}(3^{10} - 2^{-10})}$$

$$\Rightarrow x = \frac{1 - 2^{-10}}{3^{10} - 2^{-10}}$$

109. (c) 110. (c)

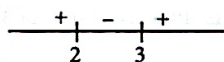
111. (b) : $f(x) = 2x^3 - 15x^2 + 36x + 1$

We have $f'(x) = 6x^2 - 30x + 36$

$$= 6(x^2 - 5x + 6) = 6(x - 3)(x - 2)$$

$$f(0) = 1$$

$$f(3) = 54 - 135 + 108 + 1 = 28$$



$$f(2) = 16 - 60 + 72 + 1 = 29$$

Then the range = (1, 29)

Hence, the given function is ONTO. But it is NOT one-one, as f' takes both positive and negative values.

112. (a)

113. (b) : The function is $f: R \rightarrow R$

$$f(x) = x^3 + 5x + 1$$

Let $y \in R$, then $y = x^3 + 5x + 1$

$$\Rightarrow x^3 + 5x + 1 - y = 0$$

As a polynomial of odd degree has always at least one real root, corresponding to any $y \in \text{co-domain}$ $\exists$ some $x \in \text{domain}$ such that $f(x) = y$. Hence f is onto.

Also f is continuous on R , because it is a polynomial function $f'(x) = 3x^2 + 5 > 0$

$\therefore f$ is strictly increasing. Hence f is one-one also.

114. (a)

$$115. (a) : F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$$

$$\Rightarrow F'(x) = 2f\left(\frac{x}{2}\right) \cdot f'\left(\frac{x}{2}\right)$$

$$+ 2g\left(\frac{x}{2}\right) \cdot g'\left(\frac{x}{2}\right) \cdot \frac{1}{2}$$

$$= f\left(\frac{x}{2}\right)f'\left(\frac{x}{2}\right) + f'\left(\frac{x}{2}\right)f''\left(\frac{x}{2}\right)$$

$$[\because g(x) = f'(x) \Rightarrow g'(x) = f''(x)]$$

$$= f\left(\frac{x}{2}\right)f'\left(\frac{x}{2}\right) - f'\left(\frac{x}{2}\right)f\left(\frac{x}{2}\right) = 0$$

$$[\because f''(x) = -f(x)]$$

$\Rightarrow F(x)$ is a constant function

$$\therefore F(x) = F(5) = 5, \forall x \Rightarrow F(10) = 5$$

116. (c) : For $x \in (-1, 1)$, we have

$$f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$\therefore f(\tan \theta) = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right)$$

(By putting $x = \tan \theta$)

$$= \tan^{-1}(\tan 2\theta) \Rightarrow f(x) = 2 \tan^{-1} x$$

$$\Rightarrow -\frac{\pi}{2} < \tan^{-1}\left(\frac{2x}{1-x^2}\right) < \frac{\pi}{2}$$

117. (b) : $(f - g)(x) = f(x) - g(x)$

$$= \begin{cases} x, & \text{if } x \text{ is rational} \\ -x, & \text{if } x \text{ is irrational} \end{cases}$$

Distinct elements of R have distinct images in $f - g$.

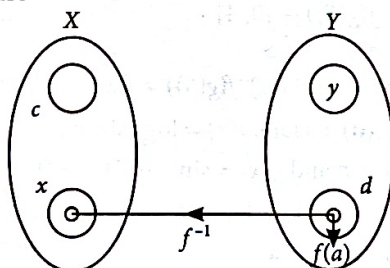
$\therefore f - g$ is one-one.

Range of $f - g$ is R .

$\therefore f$ is onto.

$\therefore f - g$ is one-one and onto.

118. (c) : The given data is shown in the figure



$$\text{Since } f^{-1}(d) = x \Rightarrow f(x) = d$$

$$\text{Now if } a \in x, f(a) \in d \Rightarrow f^{-1}(f(a)) = a$$

119. (d) : Let $f(x) = g(x) + 1$,

$$\text{where } g(x) = 2\left[\frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x\right]$$

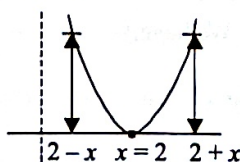
$$= 2 \sin(x - 60^\circ)$$

$$\therefore -2 \leq 2 \sin(x - 60^\circ) \leq 2$$

$$\Rightarrow -1 \leq 2 \sin(x - 60^\circ) + 1 \leq 3$$

120. (c) : If $y = f(x)$ is symmetrical about the line $x = \alpha$, then

$$f(x + \alpha) = f(x - \alpha)$$



$$\therefore f(x + 2) = f(x - 2)$$

$$121. (b) : f(x) = \frac{p(x)}{q(x)} \text{ (say)}$$

Domain of $f(x)$ is $Dp(x) \cap Dq(x)$, $q(x) \neq 0$

Now domain of $p(x)$ is $-\frac{\pi}{2} \leq \sin^{-1}(x - 3) \leq \frac{\pi}{2}$

$$\Rightarrow -\sin \frac{\pi}{2} \leq x - 3 \leq \sin \frac{\pi}{2}$$

$$\Rightarrow 2 \leq x \leq 4 \quad \dots(i)$$

$$\text{Again, } 9 - x^2 > 0 \Rightarrow x^2 < 9$$

$$|x| < 3 \text{ i.e. } -3 < x < 3 \quad \dots(ii)$$

From (i) and (ii), we have $2 \leq x < 3$

122. (b) : $g(f(x)) = (\sin x + \cos x)^2 - 1$

$= \sin 2x$ which is invertible in $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$.

123. (b)

124. (b) : Given that $f: [0, \infty) \rightarrow [0, \infty)$

such that $f(x) = \frac{x}{x+1}$

$$\text{then } f'(x) = \frac{1+x-x}{(1+x)^2} = \frac{1}{(1+x)^2} > 0 \forall x$$

$\therefore f$ is an increasing function

$\Rightarrow f$ is one-one.

Also $D_f = [0, \infty)$

$$\text{And for range let } \frac{x}{1+x} = y \Rightarrow x = \frac{y}{1-y}$$

as $x \geq 0 \Rightarrow 0 \leq y < 1$

$\therefore R_f = [0, 1) \neq \text{Codomain}$

$\therefore f$ is not onto

Hence $D_f \neq R_f \Rightarrow f$ is not onto.

$$125. (a) : \text{For } f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$$

to be defined and real $\sin^{-1} 2x + \pi/6 \geq 0$

$$\Rightarrow \sin^{-1} 2x \geq -\frac{\pi}{6} \quad \dots(1)$$

$$\text{But we know that } -\pi/2 \leq \sin^{-1} 2x \leq \pi/2 \quad \dots(2)$$

Combining (1) and (2), $-\pi/6 \leq \sin^{-1} 2x \leq \pi/2$

$$\Rightarrow \sin(-\pi/6) \leq 2x \leq \sin(\pi/2)$$

$$\Rightarrow -1/2 \leq 2x \leq 1$$

$$\Rightarrow -1/4 \leq x \leq 1/2$$

$$\therefore D_f = \left[-\frac{1}{4}, \frac{1}{2}\right]$$

126. (a) : If $y = \sin^{-1} a$, then $-1 \leq a \leq 1$

$$\therefore -1 \leq \log_3 \left(\frac{x}{3}\right) \leq 1$$

$$\left[\text{as } y = \sin^{-1} \left[\log_3 \left(\frac{x}{3} \right) \right] \right]$$

$$\Rightarrow \frac{1}{3} \leq \frac{x}{3} \leq 3^1$$

$$\Rightarrow 1 \leq x \leq 9$$

127. (a) : Given $f(x) = 2x + \sin x$

$$\Rightarrow f'(x) = 2 + \cos x$$

Since, $\cos x$ lies between -1 and 1 .

$$\Rightarrow f'(x) > 0 \quad \forall x \in \mathbb{R}$$

$\Rightarrow f(x)$ is one-one function as $f(x)$ is increasing strictly. Also, range of $f(x)$ is $(-\infty, \infty)$

Since, f is continuous and takes all values between $-\infty$ and ∞ , so f is onto.

$\Rightarrow f(x)$ is one-one and onto function.

$$128. (d) : f(x) = \begin{cases} \tan^{-1} x, & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & \text{if } |x| > 1 \end{cases}$$

$$= \begin{cases} \frac{1}{2}(-x-1), & -\infty < x < -1 \\ \tan^{-1} x, & -1 \leq x \leq 1 \\ \frac{1}{2}(x-1), & 1 < x < +\infty \end{cases}$$

$\Rightarrow f(x)$ is discontinuous at $x = -1$

and $x = 1$, domain of $f'(x)$ is $\mathbb{R} - \{-1, 1\}$.

129. (a)

130. (a) : Since, $(f \circ f^{-1})(x) = x$, we get

$$f[f^{-1}(x)] = x$$

$$\Rightarrow f^{-1}(x) + \frac{1}{f^{-1}(x)} = x$$

$$\Rightarrow (f^{-1}(x))^2 - x f^{-1}(x) + 1 = 0$$

$$f^{-1}(x) = \frac{x \pm \sqrt{x^2 - 4}}{2}$$

$$\therefore f = [1, \infty) \rightarrow [2, \infty)$$

$$f^{-1} : [2, \infty) \rightarrow [1, \infty)$$

$$\therefore f^{-1}(x) = \frac{1}{2} \left(x + \sqrt{x^2 - 4} \right)$$

$$131. (b) : 1 + x - [x] < 0$$

or $1 + x - [x] = 0$ are impossible as

$$\therefore 1 + x - [x] > 0 \quad \forall x \in \mathbb{R}$$

$$\text{Now, } f(g(x)) = f(1 + x - [x]) = 1 \quad \forall x \in \mathbb{R}.$$

132. (d)

$$133. (c) : \text{Let } h(x) = |x|,$$

$$\text{then } g(x) = |f(x)| = h(f(x))$$

Since, composition of two continuous functions is continuous, g is continuous if f is continuous.

$$134. (b) : \text{Let } y = 2^{x(x-1)}$$

$$\text{where } y \geq 1 \text{ as } x \geq 1$$

$$\log_2 y = x(x-1)$$

$$x^2 - x - \log_2 y = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 + 4 \log_2 y}}{2}$$

$$\text{For } y \geq 1, 1 - \sqrt{1 + 4 \log_2 y} \leq 0$$

$$\text{As } x \geq 1, x = \frac{1}{2} (1 + \sqrt{1 + 4 \log_2 y})$$

$$\therefore f^{-1}(x) = \frac{1}{2} (1 + \sqrt{1 + 4 \log_2 x})$$

135. (d) : Given that,

$$f(x) = \frac{x^2 - 1}{x^2 + 1} = 1 - \frac{2}{x^2 + 1}$$

$\therefore f(x)$ will become minimum when $\frac{2}{x^2 + 1}$ is maximum i.e. when $x^2 + 1$ is

minimum. i.e. at $x = 0$.

$\therefore$ Minimum value of $f(x)$ is $f(0) = -1$.

136. (b) : Given that $f(x) = 3x - 5 = y$

$$3x - 5 = y \Rightarrow x = \frac{y+5}{3}$$

$$\therefore f^{-1}(y) = \frac{y+5}{3} \text{ and } f^{-1}(x) = \frac{x+5}{3}$$

137. (a)

138. (b) : Maximum value of

$f(x) = \cos x + \cos(\sqrt{2}x)$ is 2 which occurs at $x = 0$.

Also, there is no values of x for which this value will be attained again.

139. (c) : Let $f(x) = y \Rightarrow (x+1)^2 - 1 = y$

$$\Rightarrow x = \sqrt{y+1} - 1 \quad (+ve \text{ } \angle x \geq -1)$$

$$\therefore f^{-1}(x) = \sqrt{x+1} - 1$$

$$\text{Now, } f(x) = f^{-1}(x) \quad (\text{Given})$$

$$\Rightarrow (x+1)^2 - 1 = \sqrt{x+1} - 1$$

$$\Rightarrow (x+1)^4 = x+1$$

$$\Rightarrow x = -1 \text{ or } (x+1)^3 = 1 \Rightarrow x = 0$$

$\therefore$ Required set is $\{0, -1\}$.

140. (c) : $f(x) = |px - q| + r|x|$

$$= \begin{cases} -px + q - rx, & x \leq 0 \\ -px + q + rx, & 0 < x \leq q/p \\ px - q + rx, & q/p < x \end{cases}$$

For $r = p, f'(x) < 0$ if $x \leq 0$

$$= 0 \text{ if } 0 < x \leq q/p$$

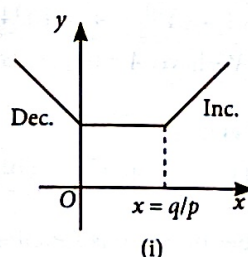
$$> 0 \text{ if } x > q/p$$

From graph (i), there are infinitely many points for min. value of $f(x)$

For $r < p, f'(x) < 0$ if $x \leq 0$

$$< 0 \text{ if } 0 < x \leq q/p$$

$$> 0 \text{ if } x > q/p$$

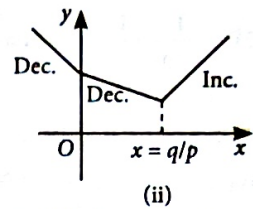


(i)

From graph (ii), there is only one point of min. of $f(x)$ at $x = q/p$

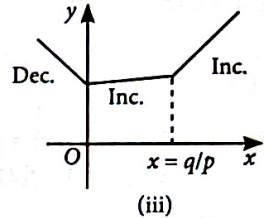
For $r > p, f'(x) < 0$ if $x \leq 0$

$$f'(x) > 0 \text{ if } x > 0$$



(ii)

From graph (iii), there is only one point of min. of $f(x)$ at $x = 0$.



(iii)

141. (d) : $f(x)$ is continuous and $x > 0$ and

$$f\left(\frac{x}{y}\right) = f(x) - f(y)$$

Also $f(e) = 1$

$\Rightarrow$ Clearly $f(x) = \ln x$ satisfies all these properties

$$\therefore f(x) = \ln x$$

142. (d) : We have $f \circ g(x) = f(g(x)) = \sin(\ln|x|)$

$$\therefore R_1 = \{u : -1 \leq u \leq 1\}$$

$$(-1 \leq \sin \theta \leq 1, \forall \theta)$$

Also, $g \circ f(x) = g(f(x)) = \ln|\sin x|$

$$\therefore R_2 = \{v : -\infty < v \leq 0\}$$

143. (a, d) :

$$f(n) = \begin{cases} K & \text{if } n = 2K - 1 \\ 2 - K & \text{if } n = 2K \end{cases} \quad \forall K \in \mathbb{N}$$

Range of $f(n) \in \mathbb{Z}$ as K and $2 - K$ cover $\mathbb{Z}$ with just $\mathbb{N}$ as inputs $\Rightarrow f$ is onto.

Observe that $f(1) = f(2) = 1 \Rightarrow f$ is not one-one.

$$g(n) = \begin{cases} 3 + 2n & \text{if } n \geq 0 \\ -2n & \text{if } n < 0 \end{cases}$$

$$\Rightarrow g \circ f(1) = g \circ f(2) \Rightarrow g \circ f \text{ is not one-one}$$

Observe that $3 + 2n \geq 3 \quad \forall n \geq 0$ and $-2n \leq -2 \quad \forall n \leq -1$

$\Rightarrow 1$ is not in range of $g(n) \Rightarrow g(n)$ is not onto.

$\Rightarrow g \circ f$ is not onto.

$$\Rightarrow f \circ g(n) = \begin{cases} \frac{g(n)+1}{2}, & g(n) \text{ is odd} \\ \frac{4-g(n)}{2}, & g(n) \text{ is even} \end{cases}$$

$$= \begin{cases} \frac{3+2n+1}{2} = n+2 \quad \forall n \geq 0 \\ \frac{4-(-2n)}{2} = n+2 \quad \forall n < 0 \end{cases}$$

$$\text{Thus } f \circ g(n) = n+2 \quad \forall n \in \mathbb{Z}$$

$\therefore f \circ g(n)$ is one-one and onto.

144. (b, c) : $f(x) = x^3 + 3x + 2, g(f(x)) = x, h(g(g(x))) = x$

$$\text{Now, } f'(x) = 3x^2 + 3$$

Again, $g'(2) = \frac{1}{f'(0)} = \frac{1}{3}$ (As we have $f(0) = 2$)

Now, $h(g(g(x))) = x$
Differentiating with respect to x ,

$$h'(g(g(x))) = \frac{1}{g'(g(x))g'(x)}$$

Now, to solve $g(g(x)) = 1$
we have $g(x) = f(1) = 6$; $f(6) = 236$

$$\therefore h'(1) = \frac{1}{g'(6)g'(236)} = \frac{1}{6 \cdot 111} = 666$$

Solving, $g(g(x)) = 0$ means $g(x) = g^{-1}(0)$.
 $\Rightarrow g(x) = 2 \therefore x = g^{-1}(2) = f(2) = 16$
 $\therefore h(0) = 16$

Again, $h(g(g(x))) = x$
Put x to $f(x)$, then $h(g(g(f(x)))) = f(x)$
 $\Rightarrow h(g(x)) = f(x) \therefore h(g(3)) = f(3) = 38$

145. (a, b, c) : We have

$$f(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)$$

As $x \in R$

$$\sin\left(\frac{\pi}{2} \sin x\right) \in [-1, 1]$$

$$\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right) \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]; f(x) \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$fog(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin\left(\frac{\pi}{2} \sin x\right)\right)\right)$$

It is observed that as $\frac{\pi}{2} \sin x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

the range of $fog(x)$ is also $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

$$\text{Again, } \lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)}{\frac{\pi}{2} \sin x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)}{\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)} \right) \cdot \frac{\pi}{2} \sin x$$

$$= \lim_{x \rightarrow 0} \frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right) = \frac{\pi}{6}$$

Again, $gof(x) = 1$

$$\Rightarrow \frac{\pi}{2} \sin\left(\sin\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right) = 1$$

$$\Rightarrow \sin\left(\sin\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right) = \frac{2}{\pi} > \frac{1}{2}$$

So, no x is possible.

$$146. (a) : f(x) = \frac{x-b}{bx-1}$$

As the range of f is not R (co-domain), hence the function is not invertible.

$$147. (a, d) : y = f(x) = \frac{x+2}{x-1}$$

$$\Rightarrow y(x-1) = x+2$$

$$\Rightarrow x(y-1) = y+2$$

$$\Rightarrow x = \frac{y+2}{y-1} \Rightarrow x = f(y)$$

Again, $f(1)$ does not exist as domain is $R - \{1\}$

$$\frac{dy}{dx} = -\frac{1}{(x-1)^2} \Rightarrow f(x) \text{ is decreasing for}$$

all $x \in R - \{1\}$

Also f is a rational function of x .

$$148. (1024) : \text{We have, } f(x) = \frac{2x}{\sqrt{1+9x^2}}$$

$$\therefore (f \circ f)(x)$$

$$= \frac{2f(x)}{\sqrt{1+9(f(x))^2}} = \frac{\frac{4x}{\sqrt{1+9x^2}}}{\sqrt{1+9 \times \frac{4x^2}{1+9x^2}}} = \frac{4x}{\sqrt{1+45x^2}}$$

$$= \frac{2^2 x}{\sqrt{1+5 \times 9x^2}}$$

$$(f \circ f \circ f)(x) = \frac{4 \times \frac{2x}{\sqrt{1+9x^2}}}{\sqrt{1+45 \times \frac{4x^2}{1+9x^2}}} = \frac{2^3 x}{\sqrt{1+21 \times 9x^2}}$$

$$(f \circ f \circ f \circ f)(x) = \frac{2^4 x}{\sqrt{1 \times 85 \times 9x^2}}$$

$$\therefore \underbrace{(f \circ f \circ f \circ \dots \circ f)}_{n \text{ times}}(x) = \frac{2^n x}{\sqrt{1+9 \left(\frac{4^n - 1}{3}\right) x^2}}$$

$$\therefore \underbrace{(f \circ f \circ f \circ \dots \circ f)}_{10 \text{ times}}(x) = \frac{2^{10} x}{\sqrt{1+9 \left(\frac{4^{10} - 1}{3}\right) x^2}}$$

$$= \frac{2^{10} x}{\sqrt{1+9 \alpha x^2}} \quad (\because \text{Given})$$

On comparing, we get

$$\alpha = \frac{4^{10} - 1}{3} \Rightarrow 3\alpha + 1 = 4^{10}$$

$$\Rightarrow \sqrt{3\alpha + 1} = \sqrt{4^{10}} = 4^5 = 1024.$$

149. (24) : We have, $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$

$$B = \{x : (x, y) \in A\}$$

$$\therefore A = \{(1, 7), (4, 5), (7, 3), (10, 1)\}$$

$$\text{and } B = \{1, 4, 7, 10\}$$

Total number of one-one functions from A to $B = 4! = 24$

150. (44) : Given, $f : A \rightarrow P(A) \Rightarrow a \in f(a)$

It means ' a ' will connect with subset which contain element a .

Total options for 1 will be 2^6 ($\because 2^6$ subsets contains 1)

Similarly, for every other element

Now, number of functions from A to $P(A)$ $= (2^6)^7 = 2^{42}$

$$\text{i.e., } m + n = 2 + 42 = 44$$

151. (180) : Total onto functions

$$= 4^5 - {}^4C_1 \times 3^5 + {}^4C_2 \times 2^5 - {}^4C_3 \times 1^5 = 240$$

Now, when $f(a) = 1$

$$\left[4 + \frac{4}{2}\right] \times \left[3 + \frac{3}{2}\right] = 24 + 36 = 60$$

So, required number of onto functions $= 240 - 60 = 180$

152. (2039) : Given, $f(x) = ax - 3$,
 $g(x) = x^b + c$ and

$$(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$$

Let $fog(x) = h^{-1}(x)$. So, $h^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$

$$\text{Let } y = \left(\frac{x-7}{2}\right)^{1/3}$$

$$\Rightarrow y^3 = \frac{x-7}{2} \Rightarrow x = 2y^3 + 7$$

$$\therefore h(x) = fog(x) = 2x^3 + 7$$

Now, $fog(x) = a(x^b + c) - 3$

$$\Rightarrow 2x^3 + 7 = a(x^b + c) - 3$$

Comparing the coefficients of like powers, we get

$$a = 2, b = 3 \text{ and } c = 5$$

So, $fog(ac) = fog(10) = 2(10)^3 + 7 = 2007$

$$g[f(x)] = (ax - 3)^b + c = (2x - 3)^3 + 5$$

$$gof(b) = (2(3) - 3)^3 + 5 = 27 + 5 = 32$$

$$\therefore fog(ac) + gof(b) = 2007 + 32 = 2039$$

153. (3240)

154. (3125) : We have,

$$f^1(x) = \frac{3x+2}{2x+3}, x \in R - \left\{-\frac{2}{3}\right\}$$

$$f^2(x) = \frac{13x+12}{12x+13}; f^3(x) = \frac{63x+62}{62x+63}$$

$$\therefore f^5(x) = \frac{1563x+1562}{1562x+1563} = \frac{ax+b}{bx+a}$$

$$\therefore a + b = 1563 + 1562 = 3125$$

155. (31) : We have,

$$2f(a) - f(b) + 3f(c) + f(d) = 0$$

$$\Rightarrow 2f(a) + 3f(c) = f(b) - f(d)$$

Also, possible values $f(a)$ can take = 0, 1, 2, 3, 4 and possible values $f(c)$ can take = 0, 1, 2, 3

$f(c)$	$f(a)$	No. of possible ways for $2f(a) + 3f(c) = f(b) - f(d)$
0	1	7 ($\angle f(b) - f(d) = 2$)
	2	5 ($\angle f(b) - f(d) = 4$)
	3	3 ($\angle f(b) - f(d) = 6$)
	4	2 ($\angle f(b) - f(d) = 8$)

1	0	6 ($\angle f(b) - f(d) = 3$)
	2	2 ($\angle f(b) - f(d) = 7$)
	3	1 ($\angle f(b) - f(d) = 9$)
2	0	3 ($\angle f(b) - f(d) = 6$)
	1	1 ($\angle f(b) - f(d) = 8$)
3	0	1 ($\angle f(b) - f(d) = 9$)

Total no. of possible ways = 31

156. (2) : We have,

$$f(x) = \left(2 \left(1 - \frac{x^{25}}{2} \right) (2 + x^{25}) \right)^{1/50}$$

$$= \left((2 - x^{25}) (2 + x^{25}) \right)^{1/50} = (4 - x^{50})^{1/50}$$

$$f(f(x)) = x, f(f(f(x))) = (4 - x^{50})^{1/50}$$

$$g(x) = (4 - x^{50})^{1/50} + x,$$

$$g(1) = (4 - 1)^{1/50} + 1 = 3^{1/50} + 1$$

$$\text{Greatest integer of } g(1) \text{ is } [g(1)] = 2$$

157. (190) : Given $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$$f(n) = \begin{cases} 2n, & \text{If } n=1, 2, 3, 4, 5 \\ 2n-11, & \text{If } n=6, 7, 8, 9, 10 \end{cases}$$

$$f(1) = 2, f(2) = 4, f(3) = 6, f(4) = 8, f(5) = 10$$

$$f(6) = 1, f(7) = 3, f(8) = 5, f(9) = 7, f(10) = 9$$

$$\text{We have, } f[g(n)] = \begin{cases} n+1, & \text{if } n \text{ is odd} \\ n-1, & \text{if } n \text{ is even} \end{cases}$$

$$f[g(10)] = 9 \Rightarrow g(10) = 10$$

$$f[g(1)] = 2 \Rightarrow g(1) = 1; f[g(2)] = 1$$

$$\Rightarrow g(2) = 6$$

$$f[g(3)] = 4 \Rightarrow g(3) = 2; f[g(4)] = 3$$

$$\Rightarrow g(4) = 7; f[g(5)] = 6 \Rightarrow g(5) = 3$$

$$\text{Now, } g(10) (g(1) + g(2) + g(3) + g(4) + g(5))$$

$$= 10[1 + 6 + 2 + 7 + 3] = 190$$

158. (37)

159. (18) : Let $f(x) = ax^2 + bx + c$ and $g(x) = dx + e$ be two real polynomials of degree 2 and 1 respectively.

$$\text{Consider, } f(g(x)) = a(dx + e)^2 + b(dx + e) + c$$

$$= a(d^2x^2 + e^2 + 2edx) + bdx + be + c$$

$$= ad^2x^2 + ae^2 + 2aedx + bdx + be + c$$

$$= ad^2x^2 + (2aed + bd)x + be + c + ae^2$$

$$= 8x^2 - 2x \quad [\text{Given}]$$

$$\Rightarrow ad^2 = 8, 2aed + bd = -2$$

$$\text{and } be + c + ae^2 = 0 \quad \dots(i)$$

$$\text{Now, consider } g(f(x)) = d(ax^2 + bx + c) + e$$

$$= adx^2 + bdx + dc + e = 4x^2 + 6x + 1$$

$$\quad [\text{Given}]$$

$$\Rightarrow ad = 4, bd = 6 \text{ and } cd + e = 1 \quad \dots(ii)$$

By solving (i) and (ii), we get

$$a = 2, b = 3, c = 1, d = 2, e = -1$$

$$\Rightarrow f(x) = 2x^2 + 3x + 1 \text{ and } g(x) = 2x - 1$$

$$\therefore f(2) = 8 + 6 + 1 \text{ and } g(2) = 3$$

$$\Rightarrow f(2) = 15 \text{ and } g(2) = 3$$

$$\therefore f(2) + g(2) = 15 + 3 = 18$$

160. (720) : Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$.

To find the number of bijective functions $f: A \rightarrow A$ such that $f(1) + f(2) + f(3) = 3$, the only possibility for this sum is $0 + 1 + 2 = 3$.

$\Rightarrow$ Elements 1, 2, 3 in the domain can be mapped with 0, 1, 2 only.

Therefore, number of bijective functions = $3! \times 5! = 6 \times 120 = 720$.

161. (19) : We have, $C = \{f: A \rightarrow B \mid 2 \in f(A) \text{ and } f \text{ is not one-one}\}$

Case-I : If $f(x) = 2 \forall x \in A$, then number of many-one functions = 1

Case-II : If $f(x) = 2$ for exactly two elements, then total number of many-one functions = ${}^3C_2 \cdot {}^3C_1 = 9$.

Case-III : If $f(x) = 2$ for exactly one element, then total number of many-one functions = ${}^3C_1 \cdot {}^3C_2 = 9$

Total number of functions = $1 + 9 + 9 = 19$.

162. (119) : The number of one-one functions = ${}^7C_5 \cdot 5! = 21 \cdot 120 = 2520 = \alpha$

For the number of onto functions, we do division into groups as follows :

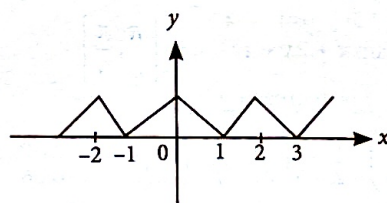
$$1 \ 1 \ 1 \ 1 \ 3 \quad \text{or} \quad 1 \ 1 \ 1 \ 2 \ 2$$

The number of onto functions is the sum in the two possible scenario, given by

$$\beta = \frac{7!}{3! \ 4!} \times 5! + \frac{7!}{2! \ 2! \ 2! \ 3!} \times 5! = 4 \cdot {}^7C_3 \times 5!$$

$$\therefore \frac{\beta - \alpha}{5!} = 4 \cdot {}^7C_3 - {}^7C_5 = 4 \times 35 - 21 = 119$$

$$163. (4) : f(x) = \begin{cases} \{x\}, & 2n-1 \leq x < 2n \\ 1 - \{x\}, & 2n \leq x < 2n+1 \end{cases}$$



We have, $f(x)$ is periodic with period

$$\int_{-10}^{10} f(x) \cos \pi x \, dx = 2 \int_0^{10} f(x) \cos \pi x \, dx$$

$$I = 2 \times 5 \int_0^2 f(x) \cos \pi x \, dx$$

$$I = 10 \left[\int_0^1 (1-x) \cos \pi x \, dx + \int_1^2 (x-1) \cos \pi x \, dx \right]$$

$$= 10 (I_1) + (I_2)$$

$$I_2 = \int_1^2 (x-1) \cos \pi x \, dx$$

$$I_2 = - \int_0^1 t \cos \pi t \, dt \quad (x-1=t)$$

We have,

$$I_1 = \int_0^1 (1-x) \cos \pi x \, dx = - \int_0^1 x \cos \pi x \, dx$$

$$I = 10 \left(-2 \int_0^1 x \cos \pi x \, dx \right)$$

$$= -20 \left[\frac{x \sin \pi x}{\pi} + \frac{\cos \pi x}{\pi^2} \right]_0^1$$

$$= -20 \left[-\frac{1}{\pi^2} - \frac{1}{\pi^2} \right] = \frac{40}{\pi^2}$$

$$\therefore \text{We have, answer} = \frac{\pi^2}{10} \times \frac{40}{\pi^2} = 4$$

164. (a) : For E_1 , the solution is given by

$$\frac{x}{x-1} > 0, \text{ which gives } x \in (-\infty, 0) \cup (1, \infty).$$

$$E_2 \text{ is given by } -1 \leq \log \frac{x}{x-1} \leq 1$$

$$\Rightarrow \frac{1}{e} \leq \frac{x}{x-1} \leq e \Rightarrow \frac{1}{e} \leq 1 + \frac{1}{x-1} \leq e$$

$$\Rightarrow \frac{1}{e} - 1 \leq \frac{1}{x-1} \leq e - 1$$

$$\Rightarrow x-1 \in \left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

$$\Rightarrow x \in \left(-\infty, \frac{1}{1-e} \right] \cup \left[\frac{e}{e-1}, \infty \right)$$

$$\text{Also, } \frac{x}{x-1} \in (0, \infty) - \{1\} \quad \forall x \in E_1$$

$$\text{Then } \log \left(\frac{x}{x-1} \right) \in ((-\infty, 0) - \{0\})$$

$$\therefore \sin^{-1} \left\{ \log \left(\frac{x}{x-1} \right) \right\} \in \left[\left[-\frac{\pi}{2}, \frac{\pi}{2} \right] - \{0\} \right\}$$

So, we have, $P \rightarrow 4, Q \rightarrow 2, R \rightarrow 1, S \rightarrow 1$

165. (i) $\rightarrow$ (B), (ii) $\rightarrow$ (C)

(i) $1 + 2x$ corresponds to (B)

(ii) $\tan x$ corresponds to (C)

166. (20) : Given, $|a - b| \geq 2$

If $(a - b) \geq 2$, then

		Total
$a = 1$	$b = 3, 4, 5, 6$	4
$a = 2$	$b = 4, 5, 6$	3
$a = 3$	$b = 5, 6$	2
$a = 4$	$b = 6$	1
		10

As $(b - a) \geq 2$ will give same number we have, total = 20

$$n(X) = {}^{20}C_6 = {}^mC_6 \quad \therefore m = 20$$

167. (36) : For $n(Y)$ we have $n(Y) = 0$, range of R has exactly one element.

Let's compute $n(Z)$

a	b
1	: 3, 4, 5, 6

- 2 : 4, 5, 6
3 : 1, 5, 6
4 : 1, 2, 6
5 : 1, 2, 3
6 : 1, 2, 3, 4

The number of relation that are function

$$= {}^4C_1 \cdot {}^3C_1 \cdot {}^3C_1 \cdot {}^3C_1 \cdot {}^4C_1 \cdot {}^4C_1$$

$$= (4 \cdot 3 \cdot 3)^2 = k^2 \therefore k = 36$$

168. (a) : The solution of $f(x) = f^{-1}(x)$ are given by $f(x) = x$, which gives $(x+1)^2 - 1 = x$
 $\Rightarrow (x+1)^2 - (x+1) = 0$
 $\Rightarrow (x+1)x = 0 \therefore x = -1, 0$

But as no co-domain of f is specified, nothing can be said about f being onto or not.

169. Let us put $y = \frac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$

$$\Rightarrow (\alpha + 6x - 8x^2)y = \alpha x^2 + 6x - 8$$

$$\Rightarrow (\alpha + 8y)x^2 + 6(1-y)x - (8 + \alpha y) = 0$$

Since x is real, $36(1-y)^2 + 4(\alpha + 8y)(8 + \alpha y) \geq 0$

$$\Rightarrow 9(1-2y+y^2) + [8\alpha + (64 + \alpha^2)y + 8\alpha y^2] \geq 0$$

$$\Rightarrow y^2(9+8\alpha) + y(46+\alpha^2) + (9+8\alpha) \geq 0 \dots (1)$$

For (1) to hold for each $y \in \mathbb{R}$, $9+8\alpha > 0$ and $(46+\alpha^2)^2 - 4(9+8\alpha)^2 \leq 0$

$$\Rightarrow \alpha > -\frac{9}{8} \text{ and } [46 + \alpha^2 - 2(9+8\alpha)]$$

$$[46 + \alpha^2 + 2(9+8\alpha)] \leq 0$$

$$\Rightarrow \alpha > -\frac{9}{8} \text{ and } (\alpha^2 - 16\alpha + 28)$$

$$(\alpha^2 + 16\alpha + 64) \leq 0$$

$$\Rightarrow \alpha > -\frac{9}{8} \text{ and } (\alpha - 2)(\alpha - 14)(\alpha + 8)^2 \leq 0$$

$$\Rightarrow \alpha > -\frac{9}{8} \text{ and } (\alpha - 2)(\alpha - 14) \leq 0$$

$$[\therefore (\alpha + 8)^2 \geq 0]$$

$$\Rightarrow \alpha > -\frac{9}{8} \text{ and } 2 \leq \alpha \leq 14 \Rightarrow 2 \leq \alpha \leq 14$$

Thus $f(x) = \frac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$ will be onto

if $2 \leq \alpha \leq 14$

When $\alpha = 3$

For $\alpha = 3$, $f(x) = \frac{3x^2 + 6x - 8}{3 + 6x - 8x^2}$

In this case $f(x) = 0$ implies,

$$3x^2 + 6x - 8 = 0$$

$$\Rightarrow x = \frac{-6 \pm \sqrt{36 + 96}}{6} = \frac{-6 \pm \sqrt{132}}{6}$$

$$= \frac{-6 \pm 2\sqrt{33}}{6} = \frac{1}{3}(-3 \pm \sqrt{33})$$

This shows that,

$$f\left[\frac{1}{3}(-3 + \sqrt{33})\right] = f\left[\frac{1}{3}(-3 - \sqrt{33})\right] = 0$$

Therefore, f is not one-to-one.

170. It gives three cases

Case (I) : Let $f(x) = 1$.

Then $f(y) \neq 1, f(z) \neq 2$ are false.

$$\Rightarrow f(x) = 1, f(y) = 1, f(z) = 2$$

This is a contradiction to f is injective.

Case (II) : Let $f(y) \neq 1$, then $f(x) = 1$,

$f(z) \neq 2$ are false

$$\Rightarrow f(z) = 2, f(x) \neq 1$$

$\therefore$ Both x and y are not mapped to 1.

So both are mapped to 3.

$\therefore f$ is not injective a contradiction.

Case (III) : Let $f(z) \neq 2$. Then $f(x) = 1$,

$f(y) \neq 1$ are both false.

$$\therefore f(x) \neq 1, f(y) = 1 \therefore f(z) = 3, f(x) = 2$$

Hence $f^{-1}(1) = y$.

171. Given that $f(x+y) = f(x)f(y) \forall x, y$

$$\therefore f(5+0) = f(5) \cdot f(0) \Rightarrow 2 = 2f(0)$$

$$\Rightarrow f(0) = 1$$

Now, given that,

$$f'(0) = 3 \Rightarrow \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 3$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(0)f(h) - f(0)}{h} = 3$$

$$[\text{Using } f(x+y) = f(x) \cdot f(y)]$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = 3 \dots (1)$$

$$[\text{Using } f(0) = 1]$$

Now, $f'(5) = \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(5)f(h) - f(5)}{h}$$

$$[\text{Using } f(x+y) = f(x)f(y)]$$

$$= \lim_{h \rightarrow 0} \frac{2f(h) - 2}{h} \quad [\text{Using } f(5) = 2]$$

$$= 2 \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = 2 \times 3 = 6 \quad [\text{Using eq. (1)}]$$

172. $f'(x) = -\sin x - 2x - 1$

$$= -(\sin x + 2x + 1) < 0 \text{ in } \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$$

$\therefore f$ is decreasing in A

$$f\left(\frac{\pi}{3}\right) = \frac{1}{2} - \frac{\pi}{3}\left(\frac{\pi}{3} + 1\right); f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{6}\left(\frac{\pi}{6} + 1\right)$$

$$\therefore f(A) = \left[\frac{1}{2} - \frac{\pi}{3}\left(1 + \frac{\pi}{3}\right), \frac{\sqrt{3}}{2} - \frac{\pi}{6}\left(\frac{\pi}{6} + 1\right)\right]$$

173. True

$$f(x) = (a - x^n)^{1/n}, a > 0, n \text{ is +ve integer}$$

$$f(f(x)) = f\left[(a - x^n)^{1/n}\right]$$

$$= \left[a - \left\{(a - x^n)^{1/n}\right\}^n\right]^{1/n} = (a - a + x^n)^{1/n} = x$$

174. True : $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$

Let us take $f(x) = 3$.

[Infact we may take any value other than 1]

Solving,

$$\frac{x^2 + 4x + 30}{x^2 - 8x + 18} = 3 \text{ we get } x = 7 \pm \sqrt{37}$$

Let $x_1 = 7 + \sqrt{37}, x_2 = 7 - \sqrt{37}$.

Then, $f(x_1) = f(x_2) = 3$. But $x_1 \neq x_2$

$\therefore f$ is not one-to-one.

175. $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right)$

$$+ \cos x \cos\left(x + \frac{\pi}{3}\right)$$

or $f(x) = \sin^2 x + \left[\sin\left(x + \frac{\pi}{3}\right)\right]^2$

$$+ \cos x \cos\left(x + \frac{\pi}{3}\right)$$

$$f(x) = \sin^2 x + \frac{1}{4}(\sin x + \sqrt{3} \cos x)^2$$

$$+ \frac{1}{2} \cos x (\cos x - \sqrt{3} \sin x)$$

$$= \frac{5}{4}(\sin^2 x + \cos^2 x) = \frac{5}{4}$$

$$(g \circ f)x = g[f(x)] = g(5/4) = 1$$

176. The total number of distinct functions from A to A is $n \times n \times \dots \times n$ (n times) i.e., n^n . The number of onto functions from A to A is

$${}^nC_0 n^n - {}^nC_1 (n-1)^n + {}^nC_2 (n-2)^n - {}^nC_3 (n-3)^n$$

$$+ \dots + (-1)^{n-1} {}^nC_{n-1} (1)^n = n!$$

177. For $f(x)$ to be defined, we should have,

$$-1 \leq \log_2 \left(\frac{x^2}{2}\right) \leq 1$$

$$\Rightarrow 2^{-1} \leq \frac{x^2}{2} \leq 2^1$$

$$\Rightarrow 1 \leq x^2 \leq 4$$

$$\Rightarrow -2 \leq x \leq -1 \text{ or } 1 \leq x \leq 2$$

$$\Rightarrow [-2, -1] \cup [1, 2]$$

178. For the given function to be defined,

$$x^2 - \frac{\pi^2}{16} \leq 0 \Rightarrow \left(x + \frac{\pi}{4}\right)\left(x - \frac{\pi}{4}\right) \leq 0$$

$$\therefore D_f = \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

Now, for $x \in D_f$,

$$\sqrt{\frac{\pi^2}{16} - x^2} \in \left[0, \frac{\pi}{4}\right] \text{ and } \sin \sqrt{\frac{\pi^2}{16} - x^2}$$

function is decreasing in $[0, \pi/4]$

$$\sin \sqrt{\frac{\pi^2}{16} - x^2} \leq \sin \frac{\pi}{4} \leq \frac{1}{\sqrt{2}}$$

$$\therefore 0 = \sin 0 \leq \sin \sqrt{\frac{\pi^2}{16} - x^2}$$

$$\Rightarrow 0 \leq 3 \sin \sqrt{\frac{\pi^2}{16} - x^2} \leq \frac{3}{\sqrt{2}}$$

$$\therefore f(x) \in \left[0, \frac{3}{\sqrt{2}}\right]$$

